

RESEARCH REPORT LABBOOK

SINGLE AND DOUBLE-SLIT DIFFRACTION OF
WALKING DROPLETS IN A HYDRODYNAMIC
PILOT-WAVE SYSTEM

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BRISTOL UNI 13/04/2026

Week 1 - Notes

23 September 2025 13:42

Massimo started off by mentioning that he had students last year who did this and managed to set up the experiment and get it working. They got a first for it.

He said as its set up that's a good thing for us and we can use the set up and start making small improvements to it for example putting the bricks under the oil so the ripples don't get effected by hitting the walls and magnet use with a tape measure on the floor under the oil so there is a precise reading of the slit distance.

He mentioned that they ran out of time last so didn't get the camera working so that would be a good thing for us to get working at first.

Would be good to find a way to keep temperature of the oil consistent and make sure the volume of the oil is consistent. So not removing pieces too many times as oil sticks to it.

He mentioned then recording the oil drop walking through the slit and seeing it diffract after many oil drop recording - layering each path - 100 paths. Also using a detector line to record the diffraction pattern.

Also then he mentioned for later in the project to try it with two slits and look at the effect of the oil droplet mimicking quantum tunnelling. It should record a diffraction pattern on the wall like in the sketch below. Look at what happens when you cover one of the slits.

Also mentioned looking at John Bush who has loads of papers on this.

Lastly he mentioned that the frequency of the speaker needs to be just below the faraday frequency for the walking to occur. Once the oil drop hits the oil with the speaker the oil starts to ripple and the use of the accelerator means the ripples bounce off the wall and the oil drop will bounce off the front of each wave, walking forward.

Good website - <https://thales.mit.edu/bush/index.php/4801-2/>

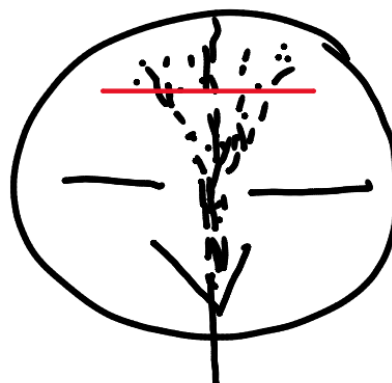
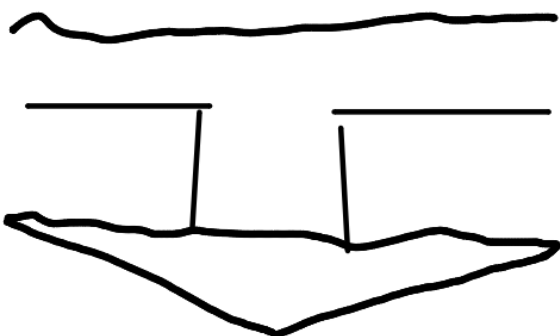
Weekly Summary (Week 1)

- Kick-off meeting with Massimo; reviewed last year's setup status and improvement areas.
- Priority set: get camera/tracking working early; improve boundary effects; keep oil volume and temperature consistent.
- Defined measurement idea: build diffraction-style distribution by stacking ~100 trajectories and sampling at a detector line.
- Longer-term: move to two-slit and test effect of blocking one slit.

Next week plan

1. Photograph/diagram full setup and write down all known parameters (bath dimensions, oil type, speaker/shaker, calibration).
2. Get camera recording stable video + decide file naming structure.
3. Measure and record slit geometry + decide detector line position.

Cross section view of the set up. Speaker below, oil and blocks to create slit.



Birdseye view of set up. Example of the oil drop movements. With the V pushing the oil drop in a straight line. The red line is the detection point.

Diffraction pattern of the oil drops at detection point with 2 slits.



Week 2 - Notes

29 September 2025

16:08

Read about Pilot wave theory and Copenhagen project

Get presentation done - most information is on blackboard

Could do 3 findings - quantum tunnelling, double and single slit diffraction, and quantum orbits

Also look through Bushes paper and greys report for certain measurements and how the experiment was set up etc.

Lab on Thursday

We went in and started looking at all the equipment which we had gone through the previous week.

We wanted to try get it up and running the first session as it was already built we just needed to put it back together.

We looked at the Arduino connected to the relay and tried sorting out where all the wires went and how the system actually worked. We connected all the wires correctly with the each wire to the correct place on the HEXFET power MOSFET as seen in the photo to the right.

We tested it many times and the relay wasn't working. We then realised that it was because of the LED light in the system, which was there to tell if the setup was working (the light would turn off every time the current was cut for a second every 10 seconds), it was taking too much current so only 2.55 V was getting to the relay rather than the required 5 V. This wasn't needed though because there was a small light on the actual Arduino already doing this.

The relay was now working with it staying up when a voltage was going through it and then it would drop on the oil and come back up for a second every 10 seconds due to the voltage being cut.

We then played around with the speaker and tried to get it vibrating. We managed to change it to the required frequency of 70Hz to get the speaker buzzing just enough that the oil wouldn't vibrate but an oil drop could bounce. The buzz was able to be felt when placing your finger on the edge of the speaker, also could be heard. We tried playing around with the amplitude but that didn't seem to do much, would be a good thing to test next time.

The main pieces of equipment were now working and we wanted to test if we could get a bouncing oil drop. We poured roughly 1mm of silicon oil onto the speaker due to not really knowing how they did it precisely so that is something to look at for a later date.

We turned on all the equipment and let it run. Oil drops were bouncing!!! But they weren't being produced from the point of the relay coming up and a drop falling off but from the relay splashing when hitting the oil. This was something to look at for next time as if this was the case it would not be a very reliable way to get one drop at a time.

We then finished for the day and were uncertain how they disposed of the oil. We kept it in there and covered it with tin foil so dust wouldn't get on the oil. As it was a test and still needed to get it working we will use this oil next time but would be good to use a new batch every time of obtaining results. We thought to dispose it by either using a syringe or some paper towels to soak it up.

We now have the setup working so for next time we will try improve the set up. Some ideas or thoughts above, as well as the ideas from week 1 for how to improve the experiment we will start to look at. We are going to set up the camera this week.

We also did a deep research using Chat GPT which came up with some references

Weekly Summary (Week 2)

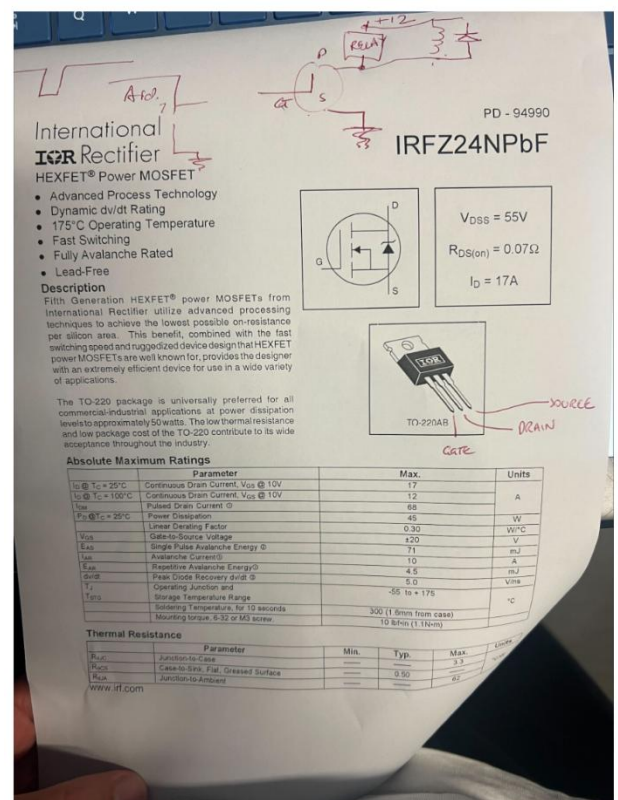
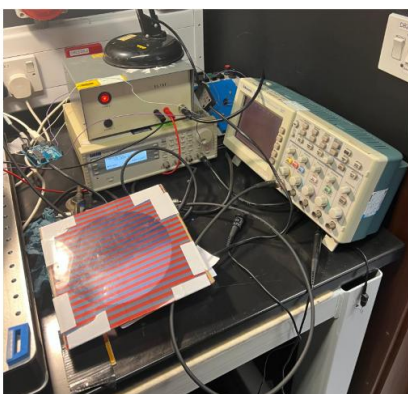
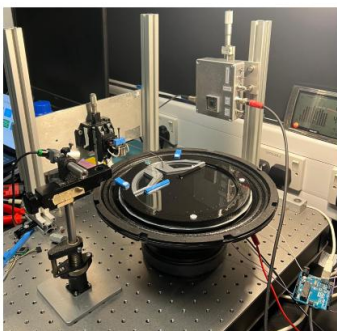
- Rebuilt and rewired Arduino-relay-MOSFET control; identified relay issue due to LED load causing low relay voltage (~2.55 V vs 5 V).
- Removed/disabled LED load → relay now actuates reliably (brief pulse ~every 10 s).
- Set driving frequency to ~70 Hz; speaker produces usable vibration/buzz.
- Added ~1 mm silicone oil and observed bouncing droplets, but droplet generation from relay tip is not yet reliable (splashing / not one droplet on demand).
- Stored oil covered (foil); need to confirm best practice for disposal / whether to use fresh oil for proper runs.

Plan going into Week 3

1. Make droplet generation repeatable (single droplet on demand, minimal splash).
2. Record and standardise operating parameters (oil viscosity + depth/volume; frequency; amplitude/drive; pulse timing).
3. Begin camera setup and calibration for tracking (stable mount + px → mm).

Success criteria

- Can produce one stable bouncing droplet for ≥ 30 s at fixed settings.
- At least one saved video where droplet + scale are clear enough to measure trajectories.



Week 3 - Notes

05 October 2025 18:09

If there are problems, look at the problems decide what they are and then find the information on how to fix them from the literature. Once that's info is found, find ways of applying it. This will save so much trial and error time.

Look at these references (in our emails with Massimo) and decide which paper gives you what information. For example thickness of oil on the speaker with that certain diameter.

Foundation of Quantum Mechanics - Travis Norsen - theory is very simple - good to read through for research.

Check pilot wave theory in comparison with the quantised orbits. Now we understand the pilot wave introduces ... how do we translate that into the dynamics of it in the oil bath.

Has anyone tried the linear movement of the quantised effects of an oil drop experiment?

Schrodinger equation plus the face equation derives the diffraction of the particles/ what guides the particle

New idea for accelerator - have mm increments on the side so you can measure the oil depth, also make it oil tight or fill in the gap so no oil seeps through.

Diameter of bath = 18.03 ± 0.01 cm

a = 9.96 ± 0.01 cm

b = 4.86 ± 0.01 cm

c = 3.43 ± 0.01 cm

d = 0.81 ± 0.01 cm

e = 2.05 ± 0.01 cm

f = 3.385 ± 0.01 cm

g = 12.1 ± 0.01 cm

Area of circle = $\pi * (18.03/2)^2 = 255.3179418155896$

Part of circle = 182.176

Small triangle = $0.81/2 * 3.385 = 1.3709$

Trapezium = $0.5 * (9.96 + 0.81) * 2.05 = 11.0393$

Area where oil is = $11.0393 + 182.176 + 1.3709 = 194.5862 \text{ cm}^2$

Therefore volume of oil to use should be $194.5862 * 0.6 = 116.7517 \text{ cm}^3$

Print a new Droplet Launcher

Tom the lab technician was working on our detector for the setup which measures the displacement of the bath. This will be used in the following weeks to calculate the acceleration of the bath.

I stayed behind and tried to play with different amplitudes and frequency's to try and get the oil drops walking. I used the readings from the report of Grace Casson, the people working on this project last year. I tried to get the depth of the oil the same as theirs but I feel like this could have been where the error lay. I found that when you increased the amplitude the oil in the bath didn't seem to react to it and my thought was that you would see some motion (ripples) in the oil. This may be because the amplifier is strong enough or the base of the speaker wasn't working properly but I wasn't too sure.

Next time I need to play around with this and try get the oil drops working. Maybe try a stronger amplifier, although the people last year found it fine. Also measure the exact amount of oil needed, not too sure about the calculations above, was a bit confused by them.

Another paper we read said 6mm but we are thinking more like 4mm. We don't want the bath too deep or the ripples won't work properly and not too shallow or the drop will hit the base.

Weekly Summary (Week 3)

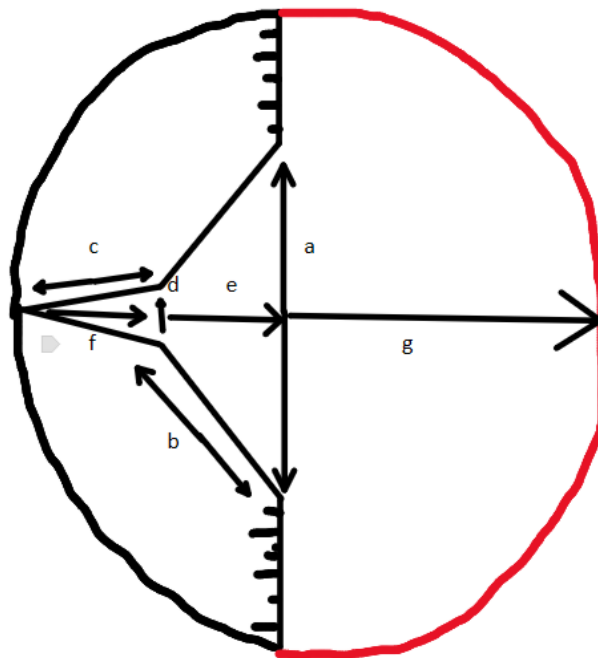
- Identified a “less trial-and-error” approach: use literature to diagnose specific setup problems (oil depth, droplet size, driving parameters).
- Measured bath geometry and estimated **oil-covered area $\approx 194.6 \text{ cm}^2$** , giving required oil volume **$\approx 116.8 \text{ cm}^3$ for 6 mm depth** (assumption noted).
- Noted improvements to make: redesign/print a new “accelerator”/insert with **mm depth markings** and better sealing to prevent seepage.
- Lab progress: tried varying frequency/amplitude to achieve walking; observed weak/unclear bath response to amplitude changes $\rightarrow$ possible amplifier/actuator coupling issue.
- Displacement detector is being worked on (Tom); planned use is to quantify bath acceleration.

Plan going into Week 4

1. **Standardise oil depth:** pick a target (e.g., 4–6 mm), measure it reliably, and record the exact oil volume used.
2. **Verify forcing properly:** confirm the shaker/speaker is actually moving the bath (measure displacement/acceleration if possible; otherwise simple diagnostic like visible Faraday ripples at higher drive).
3. **Walking attempt with documented parameters:** run a structured sweep (frequency near Faraday threshold + amplitude steps) and record what regime occurs.

Success Criteria

- Oil depth can be set and repeated to within $\sim \pm 0.5 \text{ mm}$, and volume used is recorded.
- You have at least one run where the bath response vs amplitude is *measurable* (ripples/Faraday onset or detector readout).
- A parameter table exists for each attempt (freq, amplitude setting, oil depth/viscosity, outcome).



Week 4 - Notes

13 October 2025 16:47

This week due to me being at home I downloaded Autodesk inventor and edited the new 3d printed accelerator for the set up.

I started playing around with the application and got to grips with it pretty quickly. I looked through the Lab Book of Grace Casson and found some measurements for example the diameter of the magnets used to hold it down. And then also used the measurements we had got from the last lab we did where we calculated the volume of oil needed in that space.

If we can get this printed it may work better with getting the volume of oil in the bath to a correct level without the oil seeping through into the accelerator. I did see the reason they didn't print it like this at first was because of time but that doesn't seem to be an issue on our side so we will print this.

Maybe this week/next week we try get the oil drops walking around the oil bath before doing any tests.

Harry (my lab partner) Meeting notes with Massimo -

Meeting Week 4 - 13.10.25

Monday, 13 October 2025 11:39

Theoretical problem coming from what is the assumption of the starting position. Coming from the pilot wave theory that the particle always has position, momentum etc, it must of had a probability initially and at all stages of the time evolution.

Not sure how we could change the starting position of the droplet easily, but something to consider. Add a horizontal motion to the stick, random motion by $\pm 1\text{mm}$ due to the thickness of the 'V'.

Always think about how we are going to track the trajectory of the droplet.

Suggestion was to go frame by frame and find the difference between the top and bottom half and ideally make a central point and then track that forward. Could make a software to do this or to find one that does something similar.

Talk about the experiment that I want to do - quantum tunnelling / orbits for the main part of the presentation.

What is pilot wave theory

Explain what the experiment is

Video of the apparatus us doing it, or a video form someone else about the droplet bouncing

What experiments we are going to do quantum tunnelling / orbits - have a photo of what the orbit should mirror.

Tunnelling video?

Weekly Summary (Week 4)

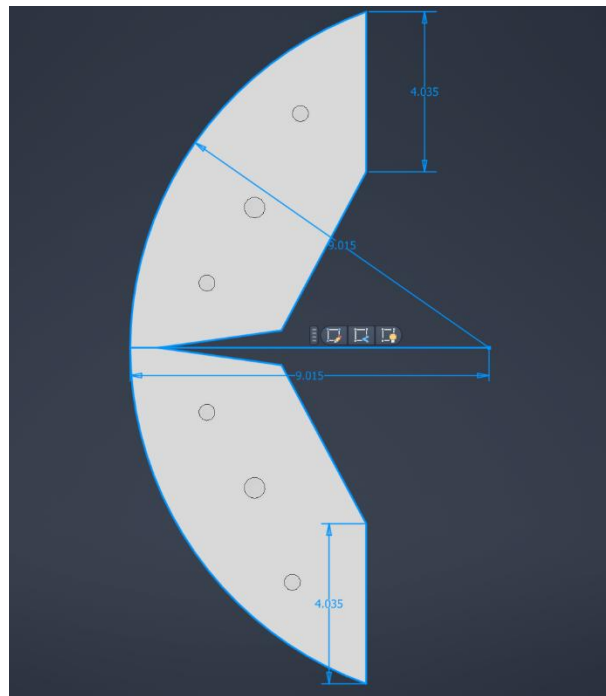
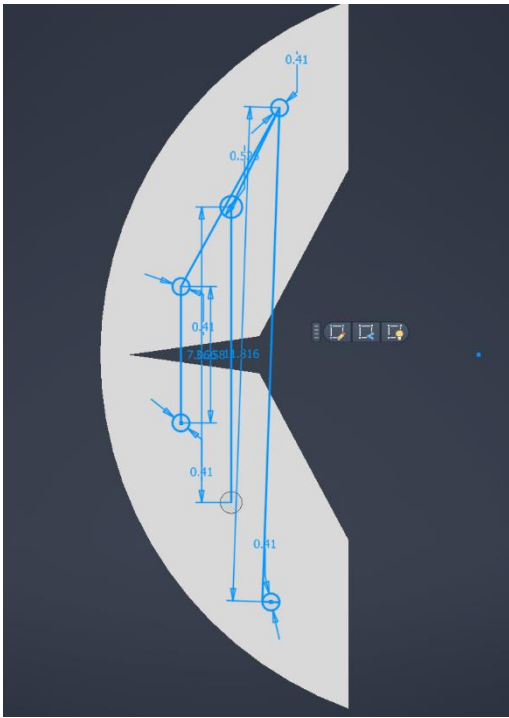
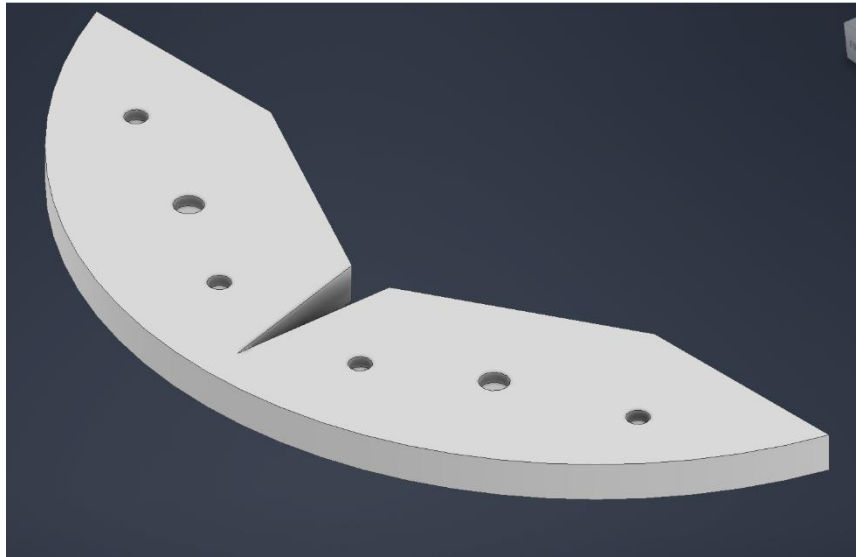
- Installed Autodesk Inventor and redesigned the 3D-printed "accelerator/insert" to improve oil containment and allow more consistent oil depth/volume.
- Used measurements from last year's notes (Grace Casson) and our own oil-volume calculations to set key dimensions (e.g., magnet hole sizing / geometry constraints).
- Concluded new design should reduce oil seepage and help standardise bath conditions; plan is to print this version.
- Meeting with Massimo: discussed importance of controlling/understanding initial conditions (droplet starting position), and prioritised trajectory tracking (frame-by-frame / software approach).
- Agreed focus for presentation: pilot-wave theory + planned experiments (tunnelling/orbits).

Plan going into Week 5

1. Get the redesigned insert **printed and fitted**; verify it seals and gives repeatable oil depth/volume.
2. Get droplets reliably **walking around the bath** before adding slit/tunnelling tests.
3. Set up trajectory tracking pipeline (even a basic version): define camera position + scaling + workflow for extracting $x(t), y(t), x'(t), y'(t)$.

Success Criteria

- Insert installed + oil depth can be set repeatably (no seepage; volume recorded).
- At least one run with stable walking captured on video suitable for tracking.



Week 5 - Notes

20 October 2025

15:21

Do the particles act like classic particles where there are parts. They just spread out but in no pattern.

To get rid of the edging effect maybe have the edges of the parts sloping. First of do a check to see that anyone has done it and see their results before trial and error.

Maybe have the bath edges sloped. This is to stop the edge effect with the slit. Similar to having them under the oil. Make a model version in fusion.

The bath has to be very flat so the oil drop doesn't walk down the slope and watch out for air currents which may cause the oil drops to drift.

<https://pages.jh.edu/rrynasi1/HealeySeminar/literature/Norsen2017FoundationsOfQuantumMechanics.pdf>

The movements of the oil droplet on an oil bath relates to some of the effects produced in pilot wave theory. In the presentation create the link between them both.

Compton frequency - de Broglie wavelength etc. all in the email he just sent:



2014_Pilot...

2014 paper very good to look at. They do all the tests of single and double slit diffraction, tunnelling and quantum orbits as well as some other tests.

Measurements:

3.65 mm small magnet

4.74 mm big magnet

Width of block - 6.45 mm

Height of block - 8 mm

Height is to be reduced to make sure block is under oil - measurement of length of block is above - reads papers to see what they used.

New height of relay for ideal drops was 6.47 mm - this may need to be adjusted each session depending on the level of the oil filled up to. Easiest way is to turn the relay off and lower it until it just touches the surface of the oil. That way it will touch every 10 seconds when turned on and a drop will form.

Recorded many videos of the oil drop working with the red and blue striped pattern and light diffuser to see the waves of the drops. During this we saw that the waves of the oil drops go quite far out so when going through the slit above the oil the waves bounce back and the oil drop has to fight through them when going through so hence why it slows down. This was discussed in our meeting with maybe making the wall sloped so there is then no bounce back.

This week was mainly focused on making the presentation which is due on Thursday the 30th

We did discuss how the detector works and after consolidation week we will take measurements of the Voltage Peak to Peak (V_{pp}) of the speaker and find the ideal amplitude for walking to occur and potentially create a graph with ideal measurements.

We also discussed what camera to use and how this will record the movement of many oil drops which we can then layer and use python to take readings and plot graphs as well as help detect the oil drops. The camera doesn't need to have a high fps as will be just be recording the oil drops from above and they move slowly any way. Just needs to be clear so you can distinctively see the oil drops.

In Week 6 consolidation week I will make the new 3d printed objects in auto desk inventor.

Weekly Summary (Week 5)

- Clarified conceptual framing for the presentation: classical-particle behaviour vs pattern formation; linked pilot-wave behaviour to quantum-style ideas (de Broglie/Compton notes + references).
- Took key hardware measurements (magnets, block dimensions) and logged an initial "best" relay height for droplet formation (~6.47 mm, dependent on oil level).
- Recorded videos using striped background/light diffuser to visualise droplet-generated waves; observed waves propagating far and interacting with the slit region (apparent slowing due to wave reflections / "fighting through" waves).
- Discussed next instrumentation steps: use displacement detector post-consolidation to measure speaker drive (V_{pp}) and find operating window for walking.
- Discussed camera requirements and using Python for tracking/plotting droplet trajectories.

Plan going into Week 6 (Consolidation week)

1. Finalise/prepare 3D-printed parts (CAD ready → print plan) and confirm block height is fully submerged during operation.
2. Set up measurement workflow for forcing: record V_{pp} and (if available) displacement/acceleration detector output vs amplitude/frequency.
3. Choose camera setup and do a basic tracking test (one short video → extract a trajectory).

Success Criteria

- A repeatable droplet-production procedure is written down (relay height procedure + settings + oil level reference).
- One dataset exists linking drive setting (e.g., V_{pp} / detector output) to observed regime (bouncing/walking/unstable).



Week 7 - Notes

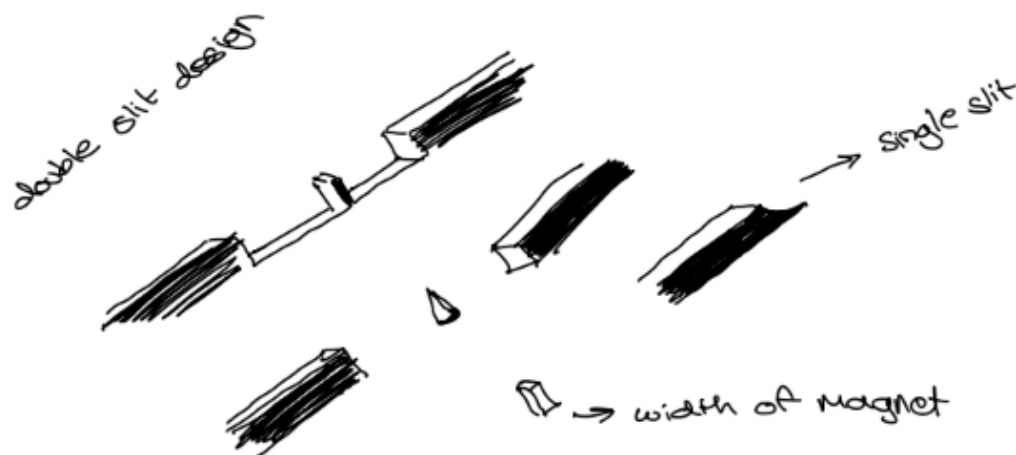
07 November 2025

15:34

In Week 7 we managed to calibrate the system. We started off by trying to figure out if the detector worked. It had 4 sections (detectors) in a square shape and essentially we attached this detector to an oscilloscope so we could read off it. The laser would reflect of a magnet that was shiny on the surface of the speaker and hit let's say the bottom of one of the sections we would then move the detector down by small increments using a micrometre and as it was rotated in the y direction the cable went from the y1-y2 direction from the detector to the oscilloscope and we would see a change in mean voltage.

We did this from the bottom of the detector till the laser went off the top of the detector measuring the mean voltage at set distances. We then plotted a graph and produced a near straight line. This enabled us to then turn on the speaker and read what the peak to peak was on the oscilloscope from the detector. We then divided this by the gradient of the slope and that gave us the vibration height of the speaker. We then worked out the error for this.

After this we sketched and took some measurements for the design of the slit. Down below are the designs we came up with. We are going to speak through these thoughts with Massimo on our meeting next Monday.



Measurements of new slit:

- Slit length : 5.2cm
- Slit width : 0.625cm
- Slit height : 0.75cm

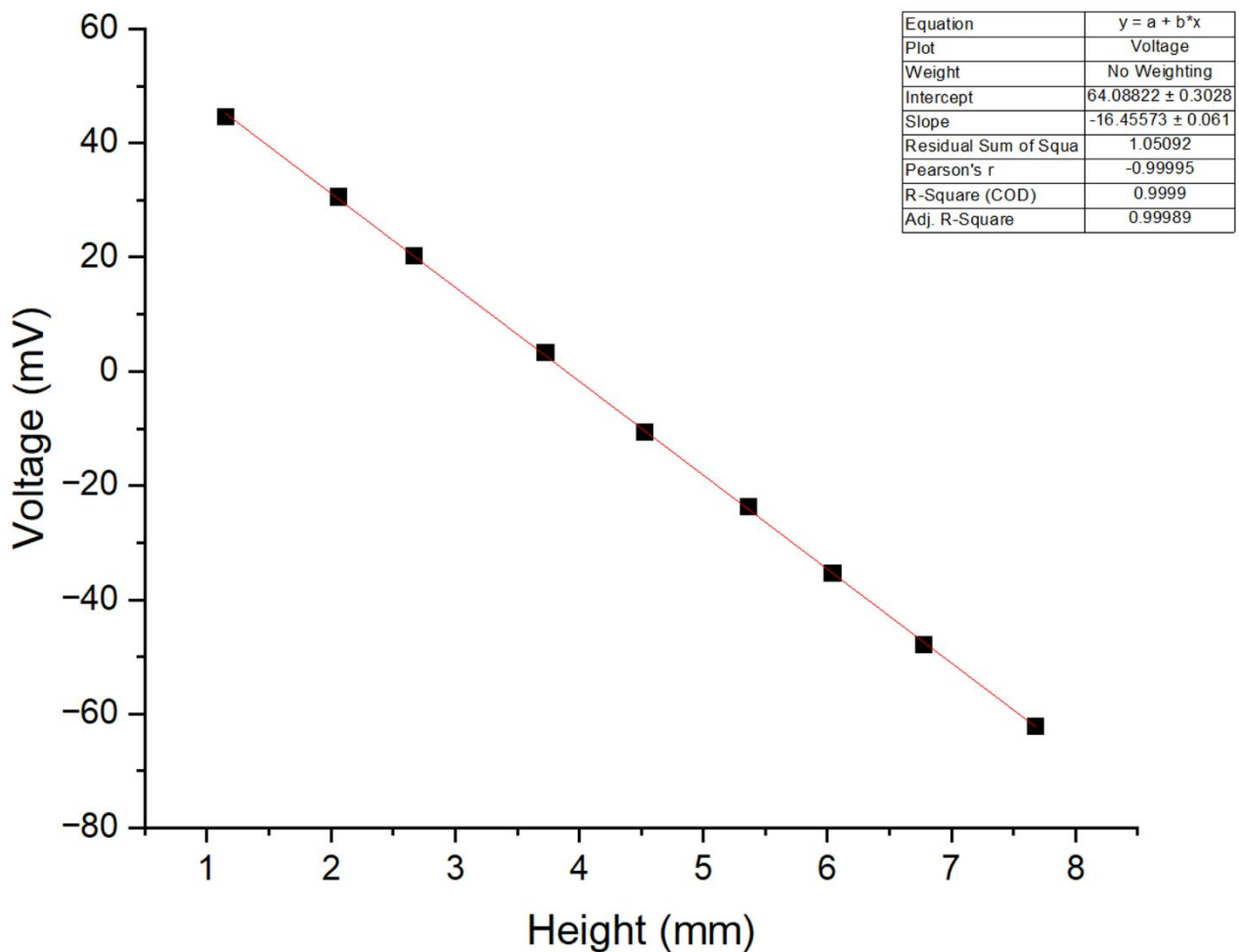
Double slit separator (cuboid)

- width : 3.65mm

Double Slit separator (cone)

- Diameter 3.65mm

	A(X)	B(Y)	C(Y)	D(Y) 	E(Y)	F(Y)
Long Name	Height	Voltage	Peak to Peak Voltage	Vibration Height	Error on Peak to Peak	Error in Vibration Height
Units	mm	mV	mV	mm		
Comments		Mean				
F(x)=				C/16.45573		
1	7.68	-62.3	16.567	1.00676	0.423	0.02597
2	6.78	-47.9				
3	6.045	-35.4				
4	5.365	-23.8				
5	4.53	-10.7				
6	3.73	3.2				
7	2.67	20.2				
8	2.06	30.5				
9	1.15	44.6				



Weekly Summary (Week 7)

- Calibrated the displacement detector by stepping detector height with a micrometer and recording mean voltage at each position.
- Plotted Voltage (mV) vs Height (mm) and obtained an approximately linear calibration gradient (mV/mm).
- Used measured peak-to-peak voltage on the oscilloscope plus the calibration gradient to estimate the speaker/bath peak-to-peak vibration height (with uncertainty).
- Sketched and took initial measurements for single- and double-slit designs; prepared to discuss design choices with Massimo next Monday.

Plan going into Week 8

- Confirm calibration is robust (repeat the height sweep once; check linear range / saturation).
- Use the detector to map vibration amplitude vs drive setting (e.g., V_{pp} vs amplifier setting / frequency) and identify the walking regime window.
- Finalise slit geometry and decide which design to build/print, based on constraints + expected wave reflections.

Success Criteria

- A repeatable calibration factor (mV/mm) with stated linear range and uncertainty.
- A simple table/plot of drive setting → vibration amplitude, with notes on droplet regime (bouncing/walking/unstable).

Week 8 - Notes

10 November 2025 16:11

Find the ratio of the slit width to the width of the slit. Use a magnet in the middle then move the outside slits to make sure the ratio is correct. The ratio is very fine tuned. Finding this will save a lot of time rather than trial and error. This is all to do with the diffraction slits.

3D print the two outside slits so they are submerged. This means the magnet and the modules will be submerged so you don't get any interference from the waves bouncing off the wall of the modules.

This week get the 3d print of the module made in week 4. And then print those two outside slits that are submerged.

Make handle for the outside slit so you're able to move it in and out to change the slit width.

Make sure we know the exact depth of the oil.

Make a measurement system so when moving the slit you know the distance:

Potentially make a disk which you lay on the bottom of the bath with a grid so you could measure the distance of the slit moved. Needs to cover the bottom so you faraday threshold stays the same.

Or make a measure tape on the module made in week 4 and measure it off that.

After our meeting I sketched a 3D printed model for the slit and disk to lie of the bottom on Autodesk inventor. The Slit looks good and is nearly ready to print, there is just one more measurement needed which is the height of the magnet.

I am going to make two models of this. One where the height is equal to one magnet and then another where its equal to the height of two magnets.

My thoughts with the disk is that it's not really relevant and can easily be done with either a thin sheet with a grid on stuck to the surface rather than a 3d printed model and or just a measurement system on the side.

I have created a handle so that we are able to adjust the position of the slit.

We printed V1 of the accelerator and found that the plastic expands whilst printing meaning the magnets didn't fit in the hole. So I edited the file again in Autodesk inventor to make the diameter and depth of the holes wider and deeper. And then I printed them using Bambu.

When measuring the widths and depth of the magnets again we realised that they didn't submerge under the oil so we have gone for some very small magnets. Let's see if they stick to the surface of the speaker strong enough.

The V2 of the accelerator and the slits are now being printed together, we will see on Thursday how they turn out and if we can start doing the double slit experiment.

In the lab on Thursday we started to test our 3d prints. They worked really well and we saw some double slit diffraction happening. Although the slits had very small magnets glued into it which were not very magnetic.

As soon as we put the box on to get rid of the air draft the drops seemed to start to move over the magnet which doesn't seem right and we think that's due to the magnet being circular or maybe there being a bit too much oil in the bath.

The amount of oil we used was 85 ml. This left quite a bit of depth over the magnet so we think this didn't affect the faraday threshold enough for the drop to not go over the magnet. We then decided to print another slit and centre piece for the magnet in the middle. We made the height of the slit and centre piece 4cm tall and then changed it so the slit would hold stronger magnets making it more stable.

New magnet measurements:

Height: 2.7 cm

Diameter: 3.8 cm

We have also got the camera set up so for next week we can start to take readings. We used the vimba software with the Prosilica gc1290 camera.

Weekly Summary (Week 8)

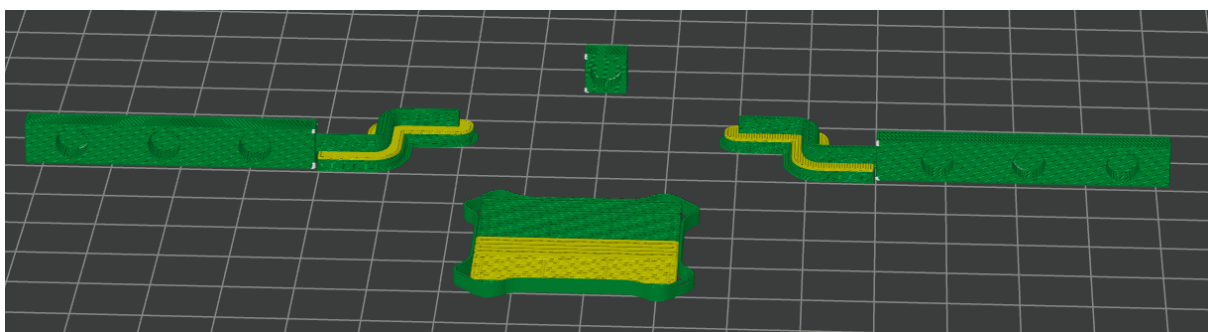
- Refined slit/accelerator design requirements: tune slit-width ratio, ensure outer slit modules are submerged, add handle, and build in a reliable measurement method for slit position/width.
- Produced CAD models for slit + optional base disk; created multiple versions to match magnet height and improve adjustability.
- Iterated 3D prints: V1 had tolerance issues (plastic expansion → magnets didn't fit); edited hole diameter/depth and printed an improved version (V2) with slits.
- Lab test: 3D printed parts worked well and we observed early signs of double-slit diffraction behaviour, but small magnets were weak / stability issues occurred (droplets drifted over magnet; likely geometry/oil depth over magnet not optimal).
- Camera setup prepared using Vimba software with Prosilica gc1290, ready for recording runs next week.

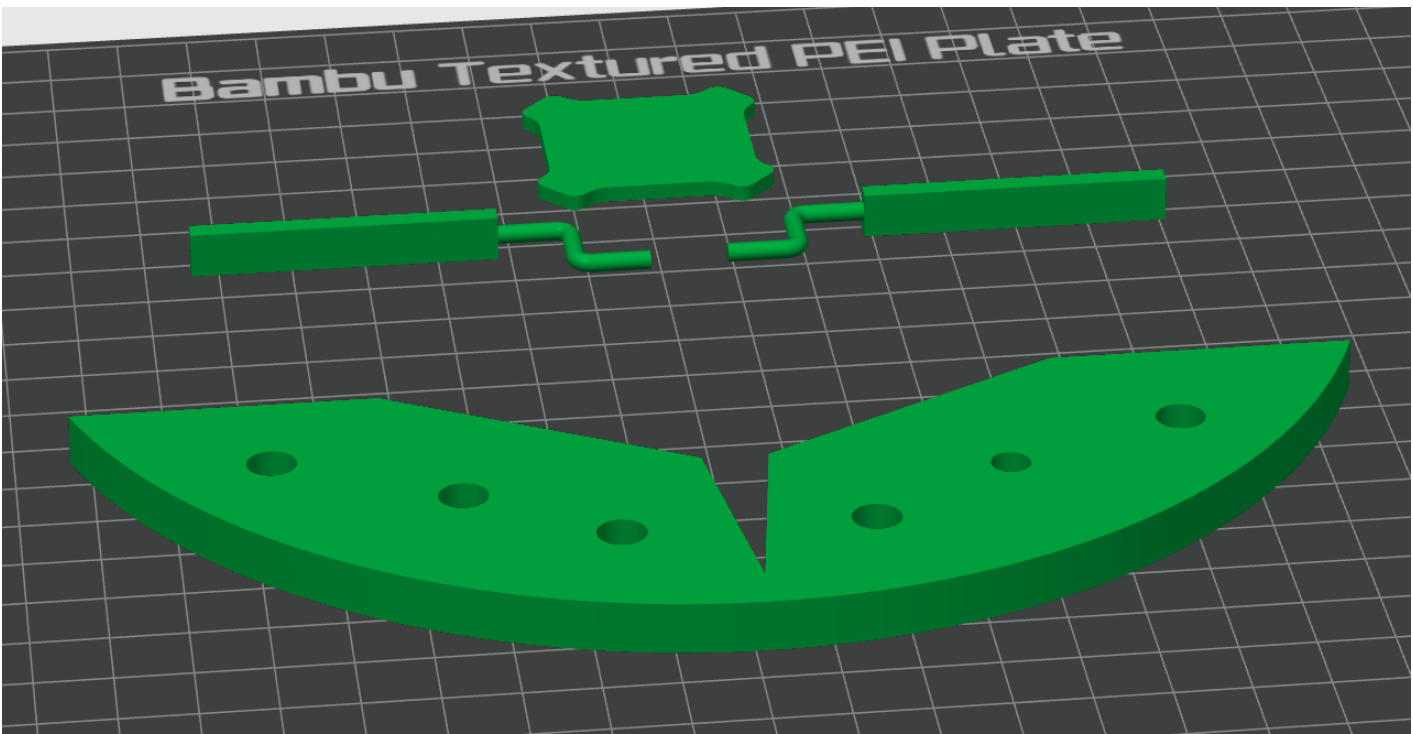
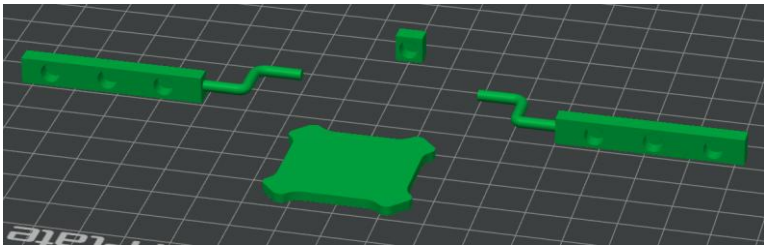
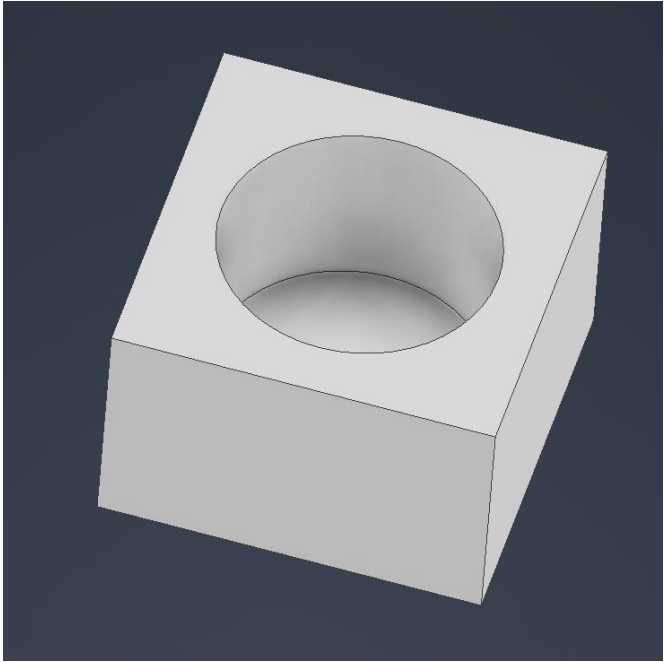
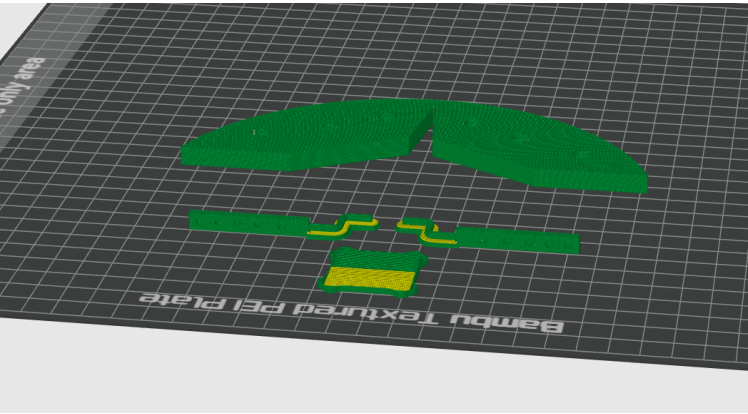
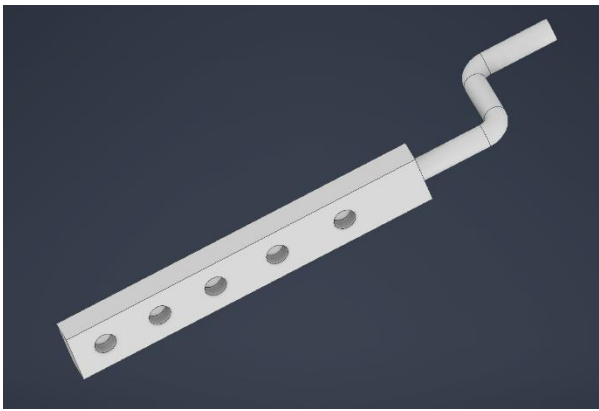
Plan going into Week 9

1. Finalise slit/magnet module so magnets are strong and reliably submerged/stable (lock in magnet size + fit).
2. Standardise bath conditions for comparisons: record oil **volume + depth**, and keep slit geometry measurable (slit width/separation + detector line reference).
3. Start structured data collection with the camera: record repeatable runs (single then double slit) and save with consistent filenames.

Success Criteria

- Slit/magnet assembly works without droplets "climbing" the magnet region unintentionally, under the same bath settings.
- At least one clean video set where trajectories can be tracked and compared across runs (same geometry + oil depth).





Week 9 - Notes

27 November 2025

16:35

Was ill at home but asked the lab technician if we could order the new magnets.

Week 10 - Notes

27 November 2025 16:36

We came in and received the new magnets we ordered.

We set up the camera attached to a clamp on a stand so it was filming above the bath. We connected the camera to the computer and opened up Synder to write the code on.

The hole at the top of our box was drilled so the camera could fit in it.

Chat GPT helped us write the code and got the camera linked to python. The camera would try pick up the droplet but it was really hard to detect and even if it did detect it, it would stop pretty soon after. We turned the lights off in the room and turned on the Lamp. We found having the lamp at the end of the setup lit up the droplets the best.

Get chat to explain the code. - Ask harry for the code.

We did get some paths from the droplets and had it plot onto a graph. It was detecting the droplet but as one droplet rather than multiple different ones firing out. We change the code and got it to pick up the path of each droplet and it plotted correctly just the drop would not pick up very often. This was really cool to see it working but we needed consistency.

We realised from reading papers that they used LED lights around the edge of the bath. We are going to implement some LED lights around the edge and this will hopefully make the drop more visible so it can read the path of each drop.

We've decided to record single slit diffraction first and then we will go onto double slit. We also thought we may plot graph A in figure 1 so we can figure out our faraday frequency and find out what the exact slit of the width should be.

We also have found a paper which is very useful for the single and double slit diffraction of bouncing oil drops so we will be referencing from here quite a bit.

We have now temporarily added led lights around the edge of the bath. This does light up the drop significantly but we are still having problems picking up the drop.

Next week we need to figure out how to fix the code. Suggestion was to only detect something on the screen if it has moved more than 3 pixels.



Pucci-Slits-2017

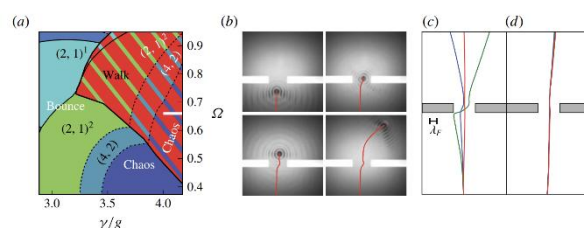


FIGURE 1. (a) Regime diagram indicating the dependence of the drop bouncing or walking state on the forcing amplitude γ/g and the vibration number $\Omega = 2\pi f \sqrt{\rho D^3 / 8\sigma}$ (Wind-Willassen *et al.* 2013; Bush 2015). In the (m, n) bouncing or walking mode, the drop bounces n times in m driving periods. The range of walkers considered in this work, of diameter $D = 0.67$ mm, is indicated by the white line as arising entirely within the chaotic walking regime. (b) Visualization of the wave field of a

Weekly Summary (Week 10)

- Received and installed new magnets; continued improving the slit/guide setup readiness.
- Mounted overhead camera (box aperture + clamp stand) and connected to PC; started Python tracking in Spyder and confirmed we can acquire frames and detect droplets intermittently.
- Main issue is **inconsistent droplet detection** (tracking sometimes stops / doesn't trigger reliably), strongly dependent on lighting. Lamp placement at end of setup gave best visibility.
- Implemented temporary LED lighting around bath edge (based on literature) which improves visibility, but tracking is still unreliable.
- Decided experimental order: **single slit first**, then double slit; identified a key reference paper for slit experiments (Pucci et al. 2017).

Plan going into Week 11

1. Stabilise tracking: lock camera settings (exposure/gain/fps), improve illumination (LED ring/diffuser), and make detection robust (thresholding + movement gating e.g., $>3 \text{ px}$).
2. Create a repeatable "recording protocol" (same bath settings + same camera settings + consistent file naming).
3. Collect the first small dataset for **single-slit** runs (even 10 usable trajectories is enough to start).

Success Criteria

- Detection works for $\geq 80\%$ of runs in a 5–10 minute session (droplet tracked continuously).
- At least 10 saved trajectories through the single-slit geometry under recorded settings.

Week 13 - Notes

23 January 2026 16:04

We got back from the Christmas holidays and the aim was to get the camera tracking working properly. We managed to get it working by scanning the tracking area and blacking it out so then when anything new entered it (the drops) we were able to then track the movement across as they weren't in the original scanned image. This worked very well and since changing the code to this we haven't run into anymore issues with the camera.

We then started recording the trajectories and collected a bit of data. We then wrote some more code to get the correct plots. So now we had that working all we needed to do next was just take the measurements for single slit diffraction. The plots included the trajectories of the drops, the individual angles of deflection and a histogram of the angles of deflection.

We then found our temporary LEDs held up with tape started to fall and were unstable. So we designed a piece which we 3D printed so that we could glue the LEDs to and then glue it to the bath. This was successful and we now have some steady lighting which makes it easier to pick up the droplets. In the process of applying the LEDs to the bath we refreshed the oil and cleaned the bath to make sure everything was good before taking measurements.

The issue we are trying to resolve now is that all of the droplets seem to be going to one side. We tried fixing this by adjusting the launcher and then making sure the whole apparatus was flat using a spirit level. This still didn't seem to fix the issue so we are trying to raise the side the droplets lean towards very slowly and trying to fine tune it as we think the spirit level may not have a high enough degree of accuracy and this is causing the droplets to veer to one side. Hopefully when this issue is fixed we will be able to start taking some proper results for the single slit diffraction.

Weekly Summary (Week 13)

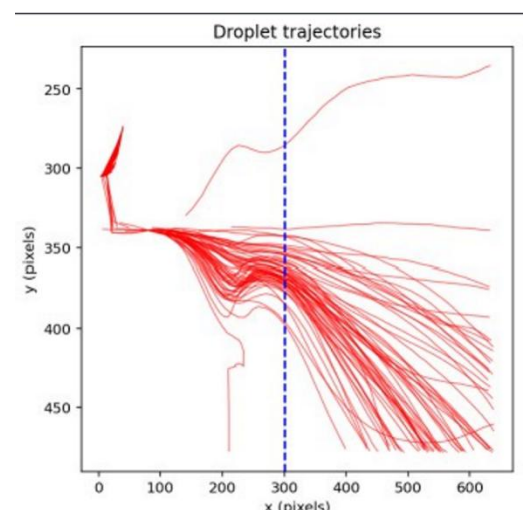
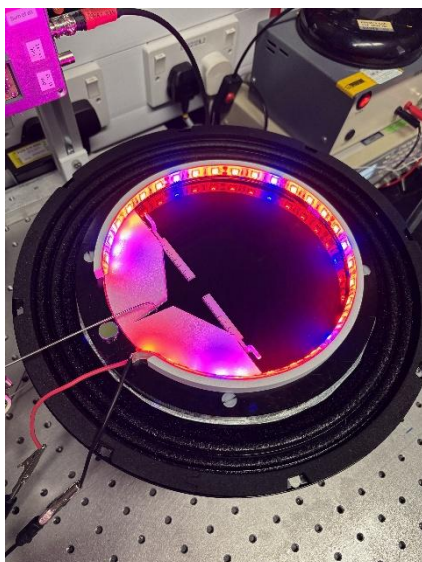
- Returned from Christmas break and resolved camera tracking: implemented background-subtraction method (scan area, black out static content, detect new objects) which eliminated previous detection issues.
- Developed plotting pipeline: trajectory overlay, per-drop deflection angles, and angular histogram — code now ready for data collection.
- Designed and 3D-printed a new LED mount to replace temporary tape-held LEDs; glued ring to bath for stable and consistent illumination. Refreshed oil and cleaned bath during installation.
- Identified persistent directional bias (all droplets drifting to one side). Attempted fixes: adjusting launcher position, levelling with spirit level, tilting bath incrementally. Issue not yet resolved — likely related to bath levelling precision.

Plan going into Week 14

1. Resolve directional bias, investigate launcher alignment, bath levelling, and airflow as possible causes.
2. Begin single-slit diffraction measurements once droplets travel straight.
3. Standardise deflection angle measurement convention and camera alignment for consistent data collection.

Success Criteria

- Droplets pass through the slit without strong systematic drift to one side.
- At least one dataset of ~50+ trajectories through a single slit with consistent parameters recorded.



Week 14 - Notes

23 January 2026 16:21

We found at 18.7 degrees C the drops seemed to not really be walking, even at the same frequency and amplitude as before. We waited for the bath to heat up via the LEDs which release a lot of heat. As soon as the bath started to heat up we saw the oil drops start to walk at the same frequency and amplitude as before. This concluded that the temperature of the bath has a big effect which aligns with the literature.

Just before taking results for single slit diffraction we still had the same problem as last week where the drops are drifting to the right every single time no matter what. This needs to be changed before taking results. We removed the slits and just kept the launcher in. We have levelled out the speaker and the stand the speaker is on and it still drifts to the right. We have done both extremes of tilting the whole thing to the left and then to the right and no matter what the drops still go to the right. We have even rotated the speaker itself, in case it was the vibrations and it still drift to the right. We have also had the box on so we don't think it is a wind current. We've been working on this all of Monday morning and just can't figure it out. We have a few more ideas we are going to try after lunch and then as soon as this is fixed we can start taking measurements.

Over the weekend I made some modules for tunnelling so when we do move onto that we will have them ready. We used the same setup as in the Predictability in a hydrodynamic pilot-wave system: Resolution of walker tunnelling paper, [Tadrist 2020](#). The width and height of the barrier is 2mm and 4.8 mm respectively, with all the other modules at 5.5 mm tall. The oil is normally at a height of 6 mm.

We changed the angle and position of the launcher and this seemed to work for some reason, the droplets were now walking straight. We decided to take some readings without the slits just to check and show that it was working, then later on in the week on Thursday we will start taking the measurements for single and double slit diffraction.

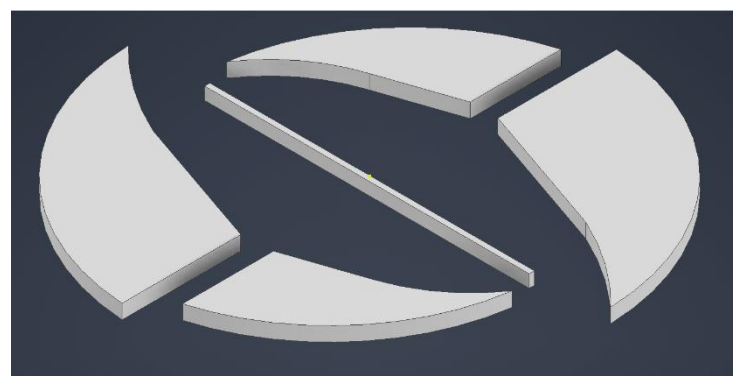
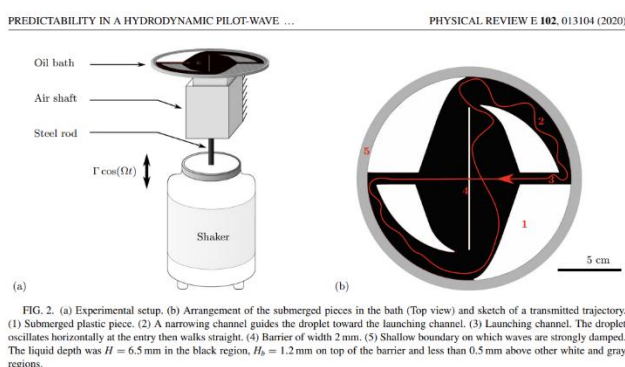
Coming in the next day we started to take measurements and they all have now started to glide to the left. We are going to try and adjust the launcher again a bit the other way. We ended up just doing very fine adjustments to the relay, the width size, the droplet launcher etc. And then we re-read the literature on Pucci 2017 slits and have concluded the insane amount of sensitivity required for this experiment. After those adjustments we found it moving to the right again and then back to the left so they had to be very finely tuned.

We got some measurements for the single slit diffraction which we were happy with as you can see in the images beside. I think we will come back to this and get more measurements later on but it's good to know that we now have it working.

We also managed to get some measurements of the oil drops going through the left slit. These showed good results and when we get some measurements for the droplets going through the other slit we are going to layer them over and you'll be able to see the interaction.

For the deflection angle measure closer together so we can get some points at 0 degrees

We changed the deflection angle so it measures from the horizontal rather than the vertical which we realised it had been doing and that was incorrect as we wanted it from the horizontal. We also implemented a vertical line in the camera imagery so that we can line up the camera at the correct position. We also marked the exact height of the camera each time to ensure all trajectory's are recorded on the same parameters.



We realised how much of an effect the level of the oil had on the drops when trying to go through the slits. If the oil was too low around the slits, the droplet would be attracted to the edge of the slits and then if the oil was too high the drops would just go over the slits and they wouldn't diffract. So the level had to be just right. Ours was set at 6 mm oil depth. And then obviously if the slits are above the oil you will get an interaction with the waves bouncing against the walls of the slits which we don't want.

We printed the modules for tunnelling and corrals but then had a chat with Massimo and realised that we may focus on just single and double slit diffraction. Getting lots and lots of results just to make sure that they are very good and there is so much to speak about if we do. Then if we have time and want to we can do tunnelling and corrals.

Lastly we have now edited the code to save all of the trajectories within an excel. This means we can layer them over one another if we have to start the recordings again because the table could have been knocked etc. Also if we want to plot another graph that we forgot about we can do that. But this had to be done with precaution and trajectories where only layered over if all parameters were the same as before. We also made the minimum trajectory pixel length 25 pixels so that got rid of a lot of bad readings. The code already has loads of rules which filter out a lot of the bad trajectories.

Amplitude was at +14.3 ~~Vpp~~ and then frequency is a 71 Hz with the slit width at 17 mm

We are then going to do a slit width at 12 mm and 22 mm

Weekly Summary (Week 14)

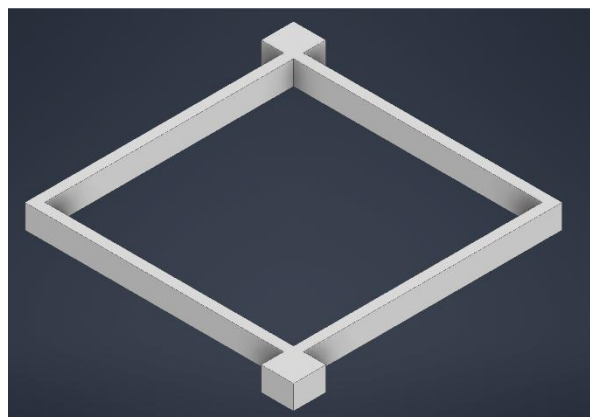
- Confirmed temperature dependence of walking: droplets did not walk at 18.7 °C but resumed at the same frequency/amplitude once the bath warmed. Consistent with literature.
- Continued troubleshooting directional bias. Tried levelling, tilting, rotating speaker, checking for air currents with box on. Concluded from Pucci et al. (2017) that the system is extremely sensitive to fine alignment.
- Changed launcher angle, which resolved the bias sufficiently to begin data collection. Took first single-slit diffraction measurements (17 mm slit) and preliminary double-slit trajectories through the left slit.
- Corrected deflection angle convention from vertical to horizontal; added vertical reference line in camera view and marked camera height for consistency across sessions.
- Identified strong sensitivity of droplet behaviour to oil depth. Too low attracts drops to slit edges, too high causes drops to pass over slits. Set oil depth at 6 mm.
- Edited tracking code to save trajectories to Excel for layering across sessions; set minimum trajectory length to 25 pixels to filter bad readings.
- Meeting with Massimo: agreed to focus on single and double slit (not tunnelling/corrals) to maximise data quality. Tunnelling modules printed but set aside.
- Parameters this session: +14.3 ~~Vpp~~, 71 Hz, 17 mm slit width. Plan to also collect 12 mm and 22 mm.

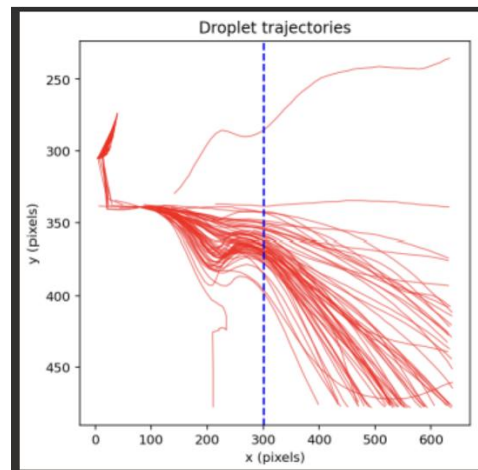
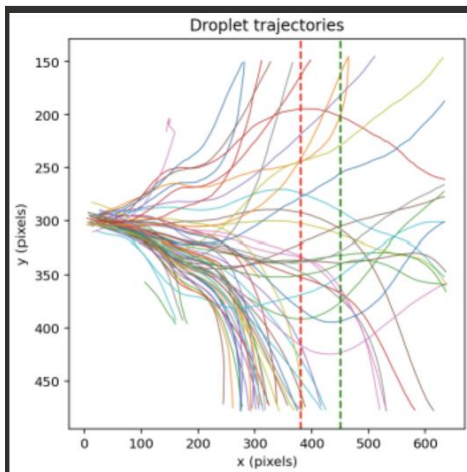
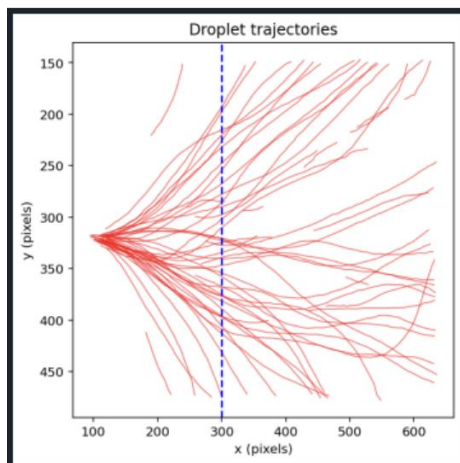
Plan going into Week 15

1. Collect single-slit data at 12 mm and 22 mm slit widths.
2. Improve speed filtering to isolate well-behaved walkers and sharpen the angular distributions.
3. Begin systematic recording of all environmental parameters (temperature, humidity) alongside each run.

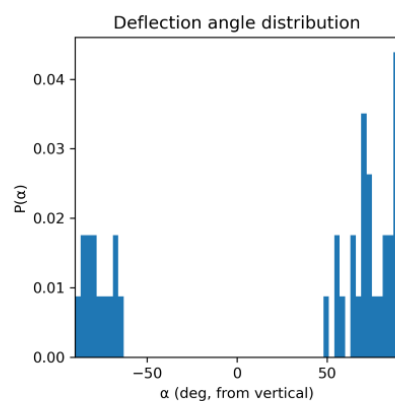
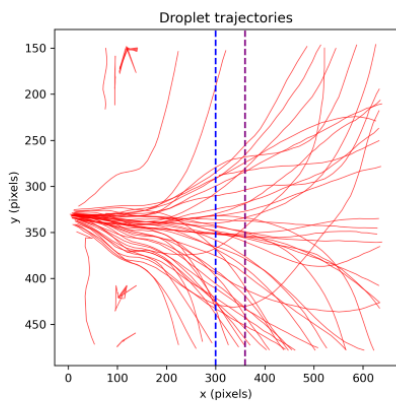
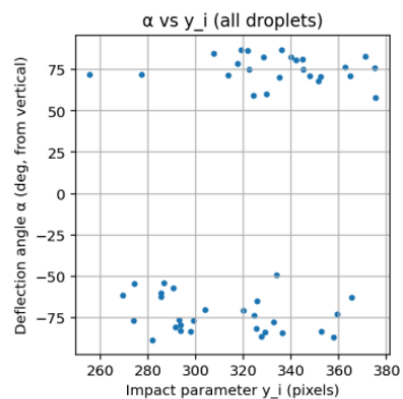
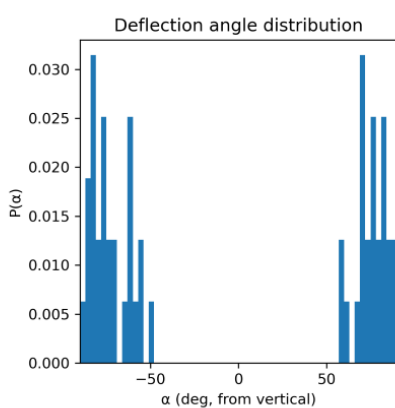
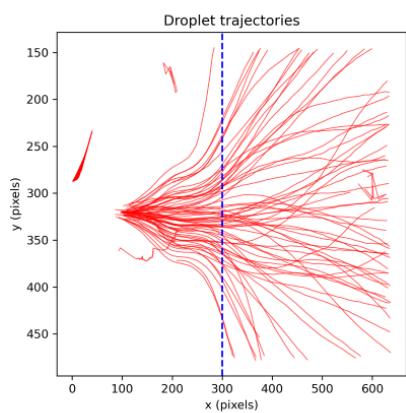
Success Criteria

- At least two additional slit widths measured with ≥ 100 trajectories each.
- Deflection angle histograms showing structure (not flat/random) for at least one slit width.

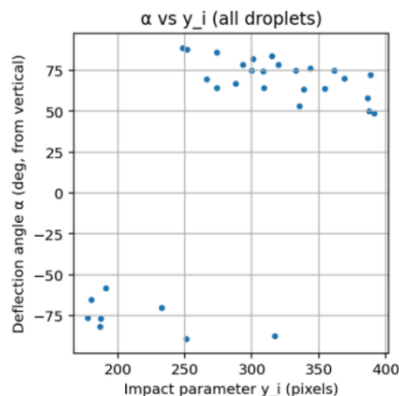
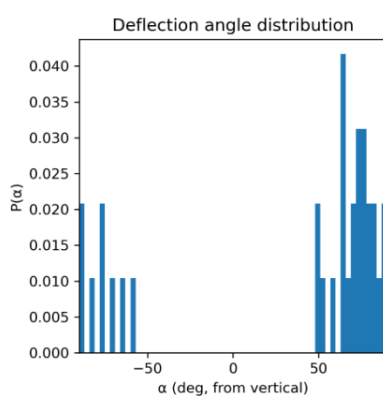
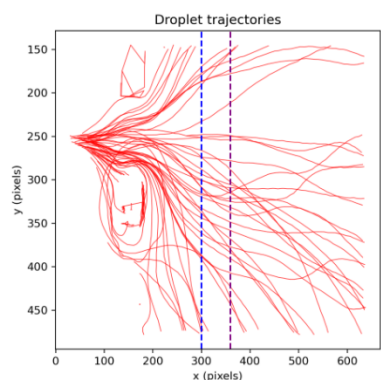




Single Slit Results



Double Slit Results



Left Slit

Week 15 - Notes

02 February 2026 14:40

Potentially keep the slits and do tunnelling by adding a curved barrier with three different sections at different widths? - Suggested by Massimo in our conversation.

Before taking the new round of recordings oil was changed this morning (2/2/26) to a height of roughly 5.5 mm $\pm$ 0.5 mm. We have now taken around 400-600 readings for each of the slit widths 12mm, 17mm and 22mm each at 70.5Hz and 14Vpp. We made sure that all of these parameters were consistent so that we could compare the results of each slit width. The widths were measured with a vernier calliper with an error of ± 0.005 mm. Only problem we found with the slit widths was getting them directly centre of the launcher as this was done by eye.

Recordings:

5/2/26, am, 12mm, 70.5Hz, 14Vpp

5/2/26, pm, 17mm, 70.5Hz, 14Vpp

6/2/26, am, 17mm, 70.5Hz, 14Vpp

6/2/26, pm, 22mm, 70.5Hz, 14Vpp

The recordings were looking good and we have been editing the code to try and analyse the readings more, plotting some different graphs taken from the literature. We realised that because the drops aren't consistent sizes this would mess around with the results but we saw a correlation between droplet size and droplet speed. And as we don't have the exact measurements of the diameter of each drop we have edited the code to track the speed of the droplets by looking at there ~~px~~px/frame. This has given us some plots which are starting to show diffraction patterns more defined as it is more finely filtered. But due to filtering this hard we lose a lot of readings meaning we had to take a lot more measurements.

We ran into a problem with our last launcher and found that the droplets where firing out a bit unreliably and there was not enough time for the droplets to straighten up and go through the slits at a constant speed. The new droplet launcher that we made was taken from Pucci's 2017 paper on slits. And since using it there has been a big effect and meant that the droplet now has more time to straighten up and go through the slits straight. We have seen a significant change in our measurements because of this.

We found that ripples where formed when the metal rod from the relay hit the oil. We had a thought that these waves where interfering with the oil drops being released. To fix this we changed the code of the Arduino so that the relay would only turn on again after 2 seconds rather than 1 second so that the ripples would have enough time to disappear before the next drop is released. We then also reduce the time in between each droplet from 10 seconds to 7 seconds so we could get more drop trajectories is the same amount of time recording.

This was the code from last year and what we edited it to.

Last year code:

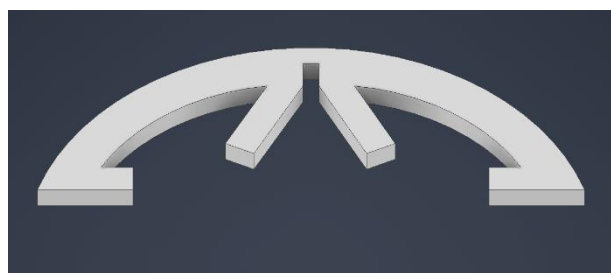
```
1  const int relayPin = 7; // Set the relay control pin
2
3  void setup() {
4      pinMode(relayPin, OUTPUT); // Set relayPin as an output
5  }
6
7  void loop() {
8      digitalWrite(relayPin, HIGH); // Turn relay ON
9      delay(500); // Wait for 0.5 seconds
10     digitalWrite(relayPin, LOW); // Turn relay OFF
11     delay(10000); // Wait for 10 seconds
12 }
```

Our code:

```
sketch_feb2a.ino
1  const int relayPin = 7; // Set the relay control pin
2
3  void setup() {
4      pinMode(relayPin, OUTPUT); // Set relayPin as an output
5  }
6
7  void loop() {
8      digitalWrite(relayPin, HIGH); // Turn relay ON
9      delay(7000); // Wait for 7 seconds
10     digitalWrite(relayPin, LOW); // Turn relay OFF
11     delay(2000); // Wait for 2 seconds
12 }
13
```

When analysing the results side by side we found that the Faraday wavelength produced by the droplet could be such that an integer multiple of it is close to the slit width, which could explain the quantised behaviour we are seeing. To test this, we plan to use slit widths at 14 mm and 19 mm to see whether this potential quantisation disappears.

We are also observing that the spread in the deflection angle does not significantly decrease as the slit width increases. This could be a because of the same quantisation effect, and if we remove this matching condition we may see a clearer change in the angular spread.



Weekly Summary (Week 15)

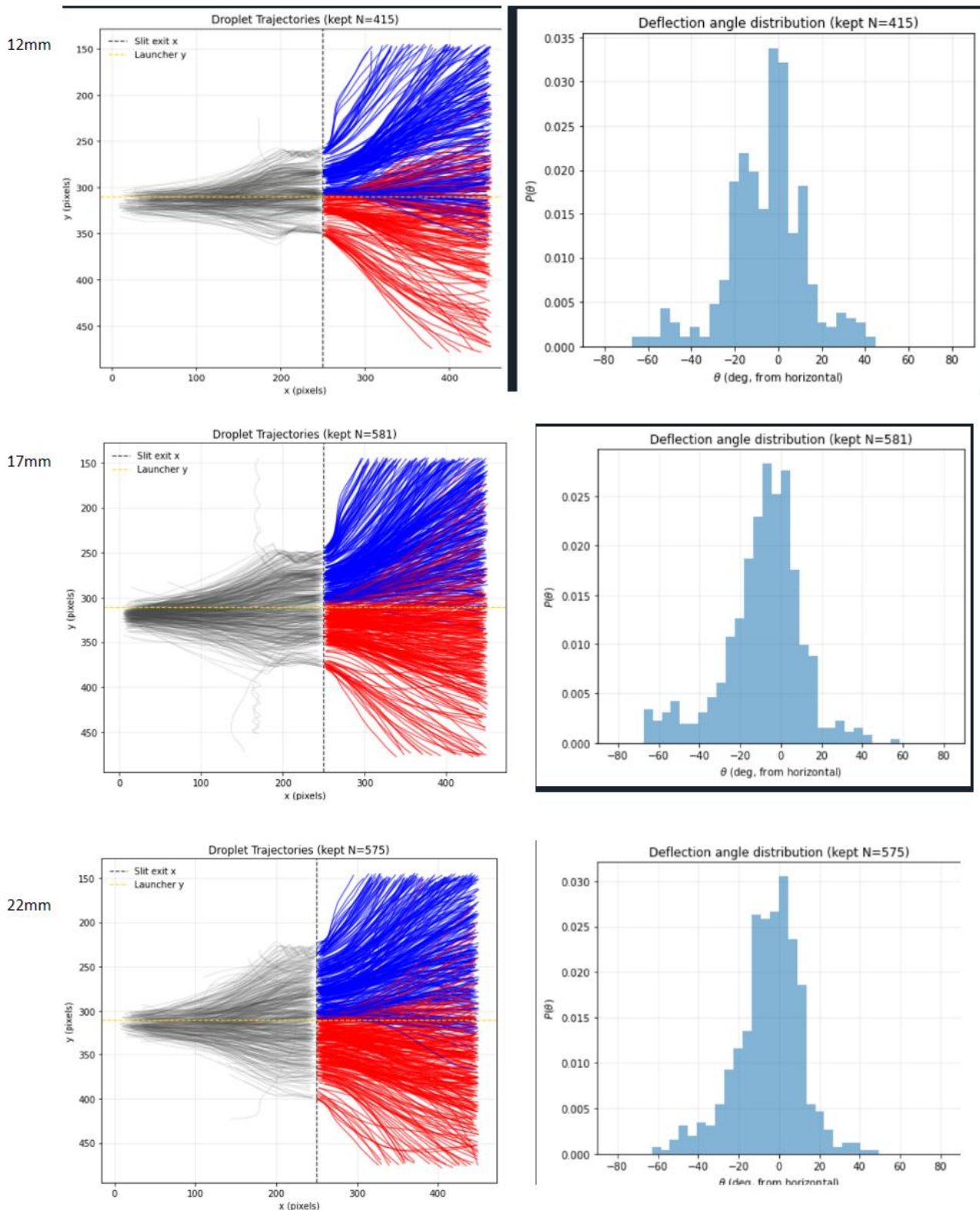
- Collected 400–600 trajectories for each of three slit widths (12, 17, 22 mm) at standardised parameters (70.5 Hz, 14.0 V_{pp}). Oil refreshed at start of session (2/2/26). Slit widths measured with vernier calliper (± 0.005 mm).
- Implemented speed-based filtering using pre-slit velocity in px/frame, which sharpened the diffraction structure in the angular distributions. Noted correlation between droplet size and speed but no direct diameter measurement available.
- Replaced the launcher with a new design based on Pucci et al. (2017) — longer channel gives the droplet more time to straighten and reach a consistent speed. Significant improvement in data quality.
- Fixed relay-induced ripple problem: changed Arduino code so relay pauses for 2 s (up from 1 s) before reactivating, and reduced inter-droplet interval from 10 s to 7 s to maintain throughput.
- Observed possible quantisation in the angular distributions when slit width is near an integer multiple of the Faraday wavelength. Plan to test at 14 mm and 19 mm where this matching condition should not hold.

Plan going into Week 16

- Measure at additional slit widths (19 mm, and potentially 14 mm) to test the wavelength-quantisation hypothesis.
- Record full environmental parameters (temperature, humidity) alongside every run for reproducibility.
- Begin comparing deflection angle distributions quantitatively to Fraunhofer sinc^2 predictions.

Success Criteria

- At least one new slit width measured with ≥ 300 trajectories under standardised conditions.
- Clear comparison between slit widths that are/aren't integer multiples of the Faraday wavelength.



Week 16 - Notes

10 February 2026 18:48

We started off this week by testing our theory about changing the slit width to a different multiple. So we set up the single slit experiment for 19mm slit width. The results were taken smoothly with not many adjustments needed. The same frequency and amplitude was set and droplet were produced consistently.

Experimental parameters:

All readings have been consistent since 2/2/26

- Ambient temperature: 26-28 degrees (ANENG GN550 IR temperature gun measured by pointing it at oil surface in the middle of bath)
- Humidity: 53+-5% (testo 623 is near the box so is not the most precise measurement)
- Air quality - good shield (plastic box) always covers the apparatus.
- Bath - 18.03cm diameter
- Levelling - measured with a spirit level
- Dwell Time - 2s

Equipment:

- Power supply for LEDs (standard LEDs) - Aim Tti EL 155R 15V 5A
- Camera (attached by clamp and stand) - Prosilica GC1290
- Detector (height controlled by a micrometer stage) - Hamamatsu Photonics S5990-01 Visible Light Position Sensing Detector (PSD), Surface Mount Ceramic.
- Relay controlled by Arduino and height controlled by nanosight micrometer stage - Made by Grace and Ben (last years project)
- Breadboard
- Speaker - Fane Sovereign 12300 12" diameter
- Power supply for speaker - EL151 Power supply connected to a Tti tg1000 10MHz DDS Function Generator
- Oscilloscope to measure bath amplitude - Tektronix TDS 2014 4 Channel Digital storage oscilloscope 100MHz 1GS/s
- Laser to measure acceleration - Thorlabs PL201 series 400-700nm
- Thorlab metal board where all apparatus was resting on

A table recording each run with all of the parameters to keep consistency.

Date	Slit width (mm)	Frequency (Hz)	Amplitude V	Temperature (degrees Celsius)	Humidity	Oil height (mm)	Submerged height (mm)	Velocity range (px/frame)
5/2/26 am	12mm	70.5Hz	14.0Vpp	28+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	8-13
5/2/26 pm	17mm	70.5Hz	14.0Vpp	28+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	8-16
6/2/26 am	17mm	70.5Hz	14.0Vpp	28+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	8-16
6/2/26 pm	22mm	70.5Hz	14.0Vpp	28+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	8-16.5
9/2/26 am	19mm	70.5Hz	14.0Vpp	28+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	
9/2/26 pm	43mm	70.5Hz	14.0Vpp	28+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	
13/2/26 all day	Double slit - 14mm each side	70.5Hz	14.0Vpp	29+-1	53+-5%	5+-0.5mm	0.5+-0.5mm	

This week includes all of the precise measurements and equipment of everything so that the experiment is reproducible.

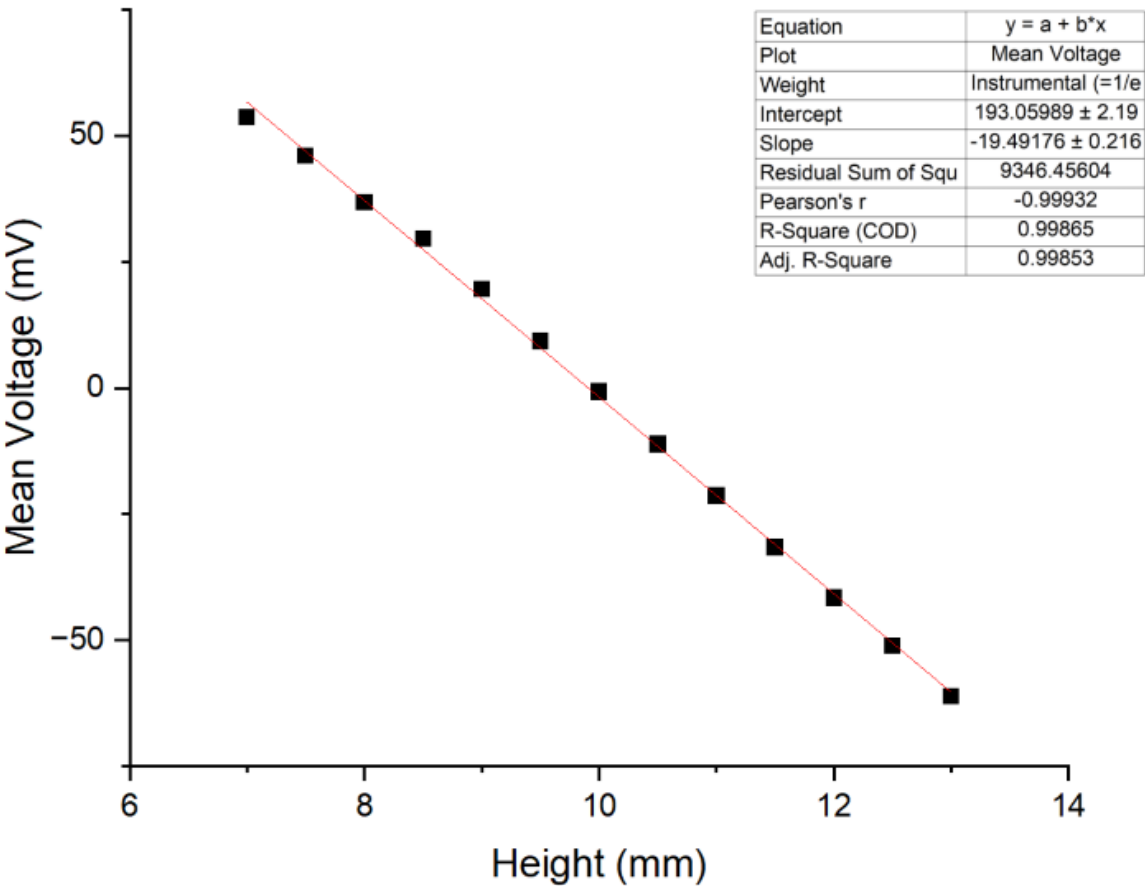
We spoke with Massimo about the quantisation we observed last week. After measuring the 19mm, it now seems clear that what we are seeing corresponds to the Fresnel (near-field) and Fraunhofer (far-field) regimes. The quantised structure within the slit is likely the Fresnel case, and the distribution further away "the deflection angle" corresponds to the Fraunhofer regime.

We also discussed the limitations of the analogue. This system only represents certain aspects of quantum mechanics, mainly wave propagation similar to that of an electron. It cannot model other properties such as spin, charge, or fields like the Higgs field. It is also not an analogue of the photon field, since photons are massless, whereas here we clearly observe a massive droplet guided by a pilot wave, which is more comparable to a massive particle such as an electron.

We then remeasured the FPS of the camera using python as the FPS we used which vimba X got us was way too high at 32.8 which then meant our droplets where moving at 40 mm/s which is way too fast. We measured the new FPS to be 9.17 which was a much more accurate reading. This meant our droplets where moving at speeds between 8 and 16 mm/s which was way more accurate. This enabled us to convert everything from px/frame into mm/s.

We decided to calibrate the bath again using the laser, detector and oscilloscope. By measuring the peak-to-peak voltage, we found that at a frequency of 70.5 Hz and an input amplitude of 14.0 V the detector reading was 15.1 mV.

When using the oscilloscope there was a lot of fluctuation so it was not very accurate when taking a measurement so we have to take an average. We then changed the oscilloscope settings and used an averaged measurement over 16 readings. This significantly reduced the observed fluctuations and produced much more stable results. Using this averaged signal, we repeated the calibration and obtained the updated results below.



Using this following equation the acceleration can be calculated:

$$\gamma = \frac{V_{pp}}{2k} \times (2\pi f)^2$$

Then using this partial derivative equation to calculate the error on γ

$$\sigma_{\gamma} = \sqrt{\left(\frac{\partial \gamma}{\partial V_{pp}} \sigma_V\right)^2 + \left(\frac{\partial \gamma}{\partial f} \sigma_f\right)^2 + \left(\frac{\partial \gamma}{\partial k} \sigma_k\right)^2}$$

The following readings were taken from the oscilloscope for the mean peak to peak voltage:

Frequency ± 0.0005 Hz	Frequency generator amplitude ± 0.05 V	Peak-to-peak voltage on oscilloscope ± 0.05 mV
70.5	14.0	17.4
70.5	15.3 (faraday threshold)	18.6

$$\gamma = \frac{17.4}{2(19.49176)} \times (2\pi \times 70.5)^2 = 87580.24747 \pm 1002.63 \text{ mms}^{-2}$$

$$\gamma_F = \frac{18.6}{2(19.49176)} \times (2\pi \times 70.5)^2 = 93620.26453 \pm 1067.55 \text{ mms}^{-2}$$

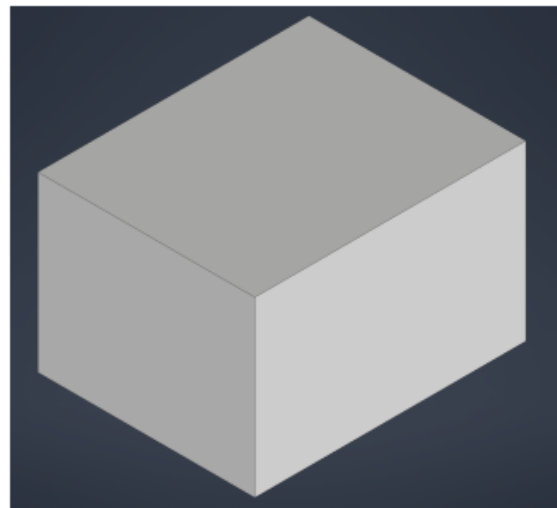
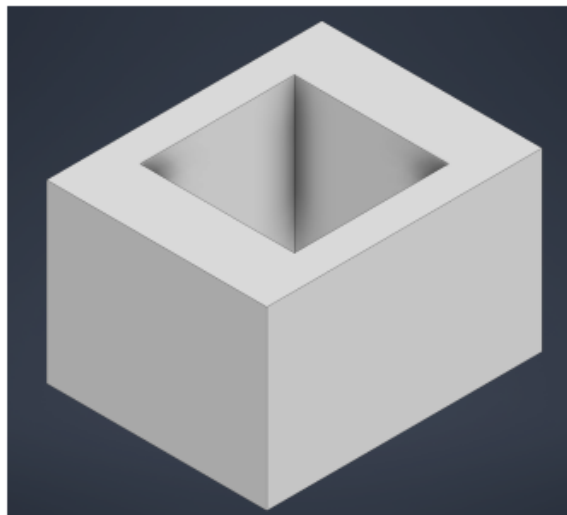
Therefore for the settings that have been used for all the individual runs, the memory parameter is

$$M = \frac{\gamma}{\gamma_F} = \frac{87580.24747}{93620.26453} = 0.9355 \pm 0.0151$$

Using this equation for the error on the memory:

$$\sigma_M = M \sqrt{\left(\frac{\sigma_{\gamma}}{\gamma}\right)^2 + \left(\frac{\sigma_{\gamma_F}}{\gamma_F}\right)^2}$$

After collecting these measurements and a load of the data for single slits we have moved onto double slit for the mean time. I 3D printed a new centre piece a lot thinner than the original piece so that the wave of the droplet could go through both of the gaps. The dimensions of the cuboid were 6.25x5x4mm. After trying this we still found that it was too big and the droplets would get stuck behind it and we would have to fire the droplets at one slit at a time which we thought would become unreliable as when trying to align the trajectories from both sides it wouldn't align properly. We wanted to fire the droplets down the middle at the centre piece and it choosing which slit to go through. So we changed the cuboid centre piece and replaced it with 3mm diameter circle magnets and of height 3mm. This meant that no trajectories would get blocked and just go around. Measurements will start being taken next week.



Weekly Summary (Week 16)

- Measured 19 mm single-slit and 43 mm no-slit control datasets. Also collected first double-slit data (14 mm each side, $d = 13$ mm separation) on 13/2/26.
- Documented full experimental parameters and equipment list for reproducibility: 70.5 Hz, 14.0 V_{pp}, 20 cSt silicone oil, 26–28 °C, 53 ± 5% humidity, 5 ± 0.5 mm oil height, 0.5 ± 0.5 mm submerged barrier height.
- Discussed results with Massimo: identified that the quantised near-slit structure corresponds to the Fresnel (near-field) regime, while the broader angular envelope corresponds to the Fraunhofer (far-field) regime. Also discussed limitations of the analogue (no spin, charge, or photon field equivalent).
- Corrected camera frame rate: Vimba X reported 32.8 fps but measured true rate is 9.17 fps using Python timing. Speeds now correctly in mm/s rather than the erroneously high values from before.
- Recalibrated displacement detector using oscilloscope averaging (16 readings). New calibration gradient $k = -19.49 \pm 0.22$ mV/mm ($R^2 = 0.999$). Calculated $\gamma/\gamma_{\text{F}} = 0.936 \pm 0.015$.
- Iterated double-slit centre piece design: cuboid separator too large (droplets stuck behind it), replaced with 3 mm diameter circle magnets which allow droplets to choose either slit.

Plan going into Week 17

1. Begin collecting double-slit trajectory data with the new centre piece magnet design.
2. Collect larger datasets for slit widths where N is still low.
3. Start comparing single-slit angular distributions quantitatively to the Fraunhofer sinc^2 envelope.

Success Criteria

- At least 200 double-slit trajectories collected with droplets passing through both slits.
- γ/γ_{F} value confirmed and documented with full error propagation.

Week 17 - Notes

16 February 2026

22:08

I unfortunately was absent this week but Harry my lab partner got some very promising results for the double slit diffraction. I had a few calls with him and discussed the results as well as did some of my own research on comparing all of the papers out there with our results. This will be spoken about in the Final Report.

At the beginning of this week I looked at some of the photos we had taken of the oil droplets and measured the wavelength between each wave. This gave me a faraday wavelength of around roughly 4.5 mm which was a lot lower than what Harry got. I did some more research and found the dispersion relation in Pilot Wave Hydrodynamics by John Bush which we then used to find our faraday wavelength. Harry ended up finding this dispersion relation too so we decided to use that as our final faraday wavelength.

This is a summary of what Harry did this week:

During this week several improvements were made to the droplet diffraction apparatus and the double-slit experiments were further analysed.

Firstly, the silicone oil in the bath was replaced on 16/02/26. Experiments were then conducted using a double slit geometry with driving frequency 70.5 Hz and amplitude around 14.0–14.1 V. The slit widths measured where 14 mm ($d = 17$ mm separation) and 10 mm ($d = 13$ mm). Environmental conditions varied slightly during the week with temperatures ranging from 19–29 °C (Roughly at 27 °C for when taking results) and humidity from 37–52%.

A persistent leftward bias in the droplet trajectories was initially observed. This was likely caused by airflow entering through the camera opening at the top of the apparatus. Covering the opening with tape (leaving a small hole for the camera) significantly reduced this bias. Some tunnelling behaviour through the Centre Piece was still occasionally observed, particularly when droplets approached the Centre Piece with higher velocities.

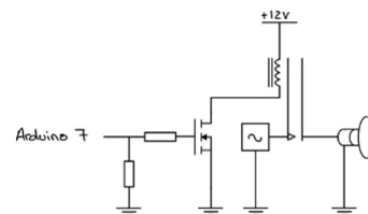
To improve trajectory control, stacked magnets were added as the Centre Piece to influence the droplet paths. Approximately 15 ml of additional oil was also added to the bath to improve the stability of the droplet motion.

Another issue identified was the build-up of droplets at the end of the bath, which could alter the local wavefield and affect subsequent droplet trajectories. To mitigate this, an automated control system was implemented using an Arduino and relay. This periodically pauses the driving signal, generates a droplet, allows it to propagate, and then resets the system. This automation allows the experiment to run for longer periods without manual intervention.

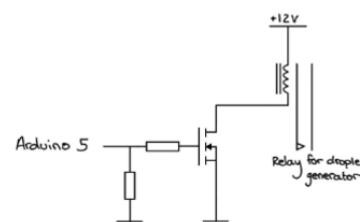
The new code that controls this:

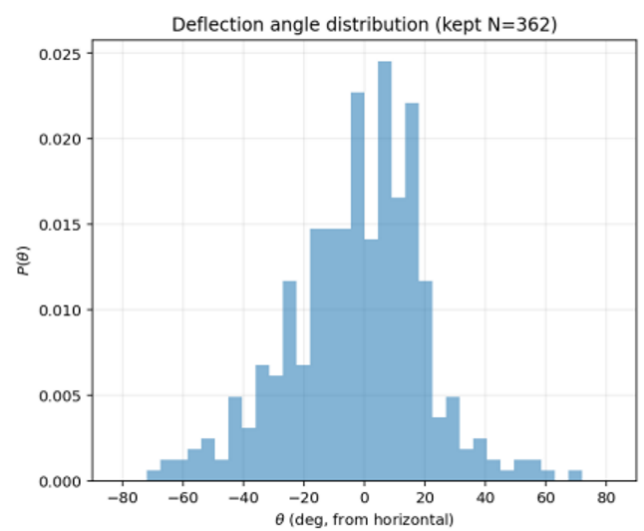
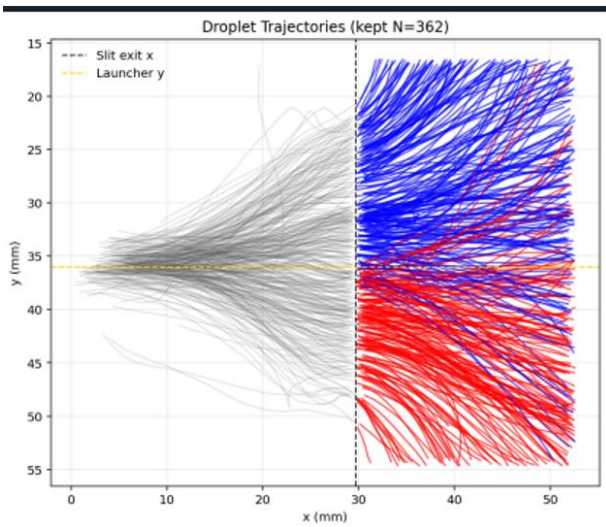
```
const int dropletPin = 7;
const int speakerDrivePin = 5;
// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
  pinMode(dropletPin, OUTPUT);
  pinMode(speakerDrivePin, OUTPUT);
  digitalWrite(LED_BUILTIN, LOW); // turn the LED on (HIGH is the voltage level)
  digitalWrite(dropletPin, LOW);
  digitalWrite(speakerDrivePin, LOW);
}
// the loop function runs over and over again forever
void loop() {
  digitalWrite(speakerDrivePin, HIGH); // activate speaker
  digitalWrite(LED_BUILTIN, HIGH);
  delay(2000);
  digitalWrite(LED_BUILTIN, LOW);
  digitalWrite(dropletPin, HIGH); // lift arm making droplet
  delay(9000); // wait for droplet to run its course
  digitalWrite(speakerDrivePin, LOW); // remove signal from speaker
  digitalWrite(dropletPin, LOW); // reset droplet arm ready for next droplet
  delay(1000);
}
```

Function Generator :



Droplet generator:





Trajectory data from the double slit experiment were compiled and analysed. Droplet speeds below approximately 9 mm s^{-1} were removed as these corresponded to smaller, less stable droplets. After filtering, several hundred trajectories were retained for analysis.

The angular deflection distributions showed a systematic bias towards positive angles, indicating an asymmetry in the system. To account for this, the distribution was reflected about the central peak to produce a symmetrised dataset, allowing comparison with theoretical diffraction predictions.

Using the measured angular distributions from single-slit experiments and fitting to the Fraunhofer envelope an estimate of the Faraday wavelength was obtained. Measurements from several slit widths produced values between $\approx 5.8 \text{ mm}$ and 8.3 mm , giving an average of $\lambda_F \approx 7.0 \pm 0.17 \text{ mm}$.

From single-slit fraunhofer, the angular envelope function is

$$I(\theta) \propto \text{sinc}^2\left(\frac{\pi a}{\lambda} \sin\theta\right)$$

And the first zero satisfies $\sin\theta_0 = \frac{\lambda}{a} \rightarrow \lambda = a \sin\theta_0$

Where a is the slit separation and the first zero is at angle θ_0 .

A summary of how a short section of code computes this angle:

The code estimates the Faraday wavelength by fitting your measured angular distribution to the theoretical single-slit Fraunhofer form. It adjusts λ (and optionally a small shift and blur) to minimise the difference between the theoretical curve and your histogram, so the fitted λ is the effective wavelength that best matches the observed diffraction envelope.

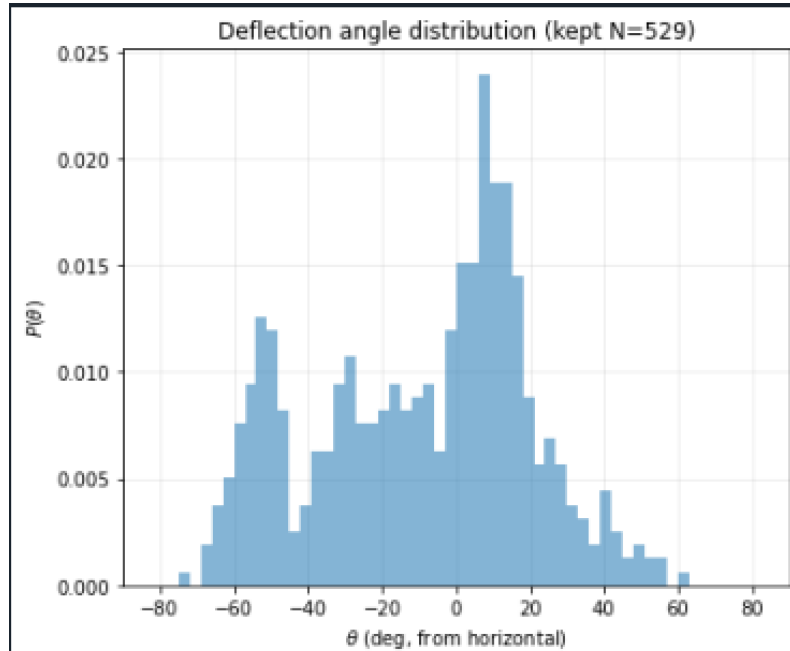
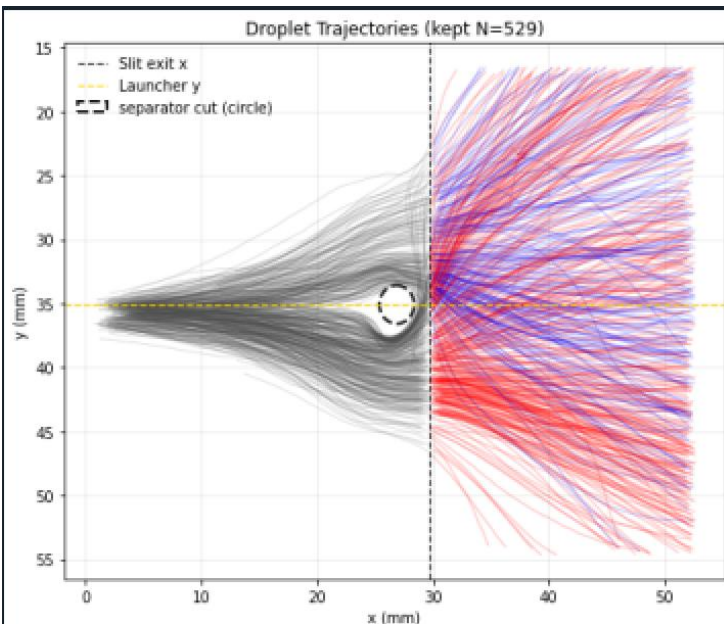
Having done this for 12, 17 and 22 mm diffraction patterns we obtained the following results for λ_F , filtering for speeds (10-14 mms^{-1}), λ_F :

12mm = 5.835mm
17mm = 8.259mm
22mm = 6.928mm

Which gives an average of $7.0073 \pm 0.1730 \text{ mm}$

Using this wavelength and the experimental slit parameters ($L = 10 \text{ mm}$, $d = 13 \text{ mm}$), the expected double-slit diffraction pattern was compared with the measured angular distribution. After centring the data, the theoretical envelope showed reasonable agreement with the experimental histogram, although some asymmetry remained due to residual experimental bias.

Overall, the week focused on improving experimental stability, automating droplet generation and disappearance, collecting double slit data, and extracting an effective Faraday wavelength from the diffraction data. Further work is required to reduce systematic biases in the apparatus and obtain more symmetric diffraction patterns.



The angle deflection histogram has a clear bias as the central peak is centred at roughly 10 degrees, if this plot is reflected at the peak this bias can be eradicated even [REDACTED] A theoretical curve can then be fitted, using the following equation and parameters:

is approximate.

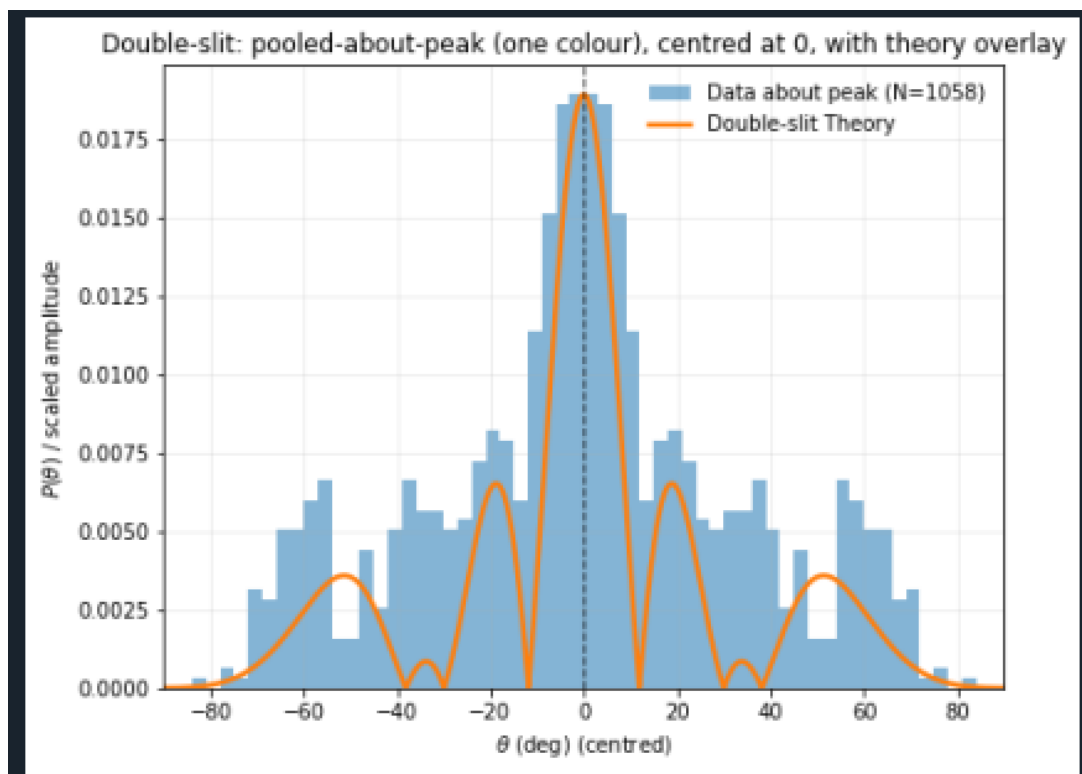
$$|A(\theta)| = \left| \text{sinc}\left(\frac{\pi L \sin \theta}{\lambda}\right) \cos\left(\frac{\pi d \sin \theta}{\lambda}\right) \right|$$

$L = 10\text{mm}$ (slit width)

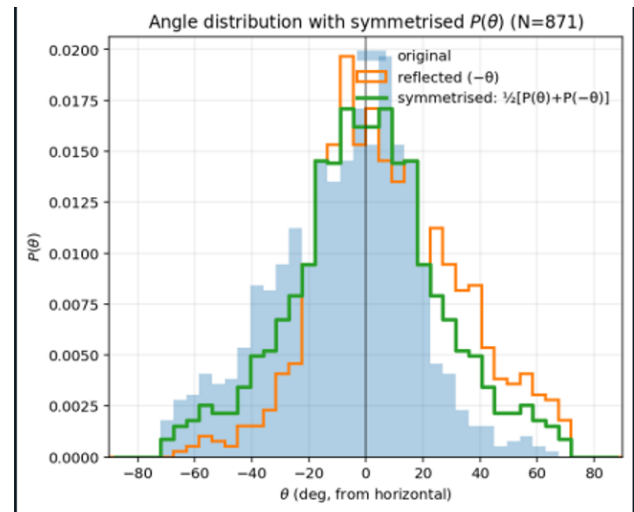
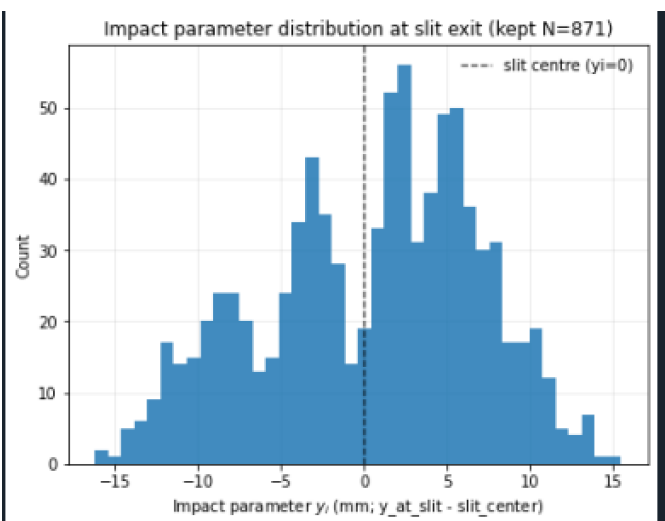
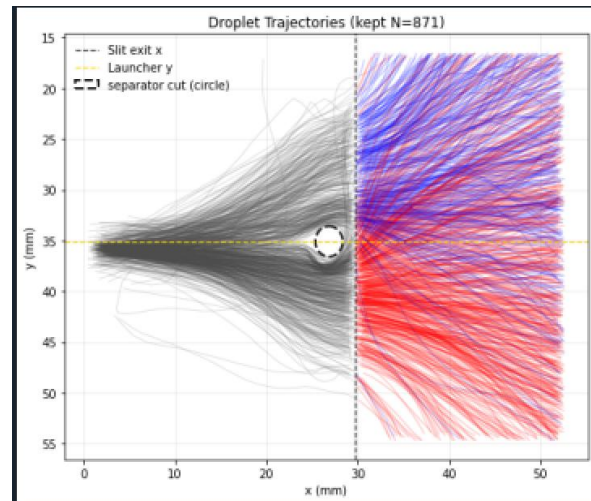
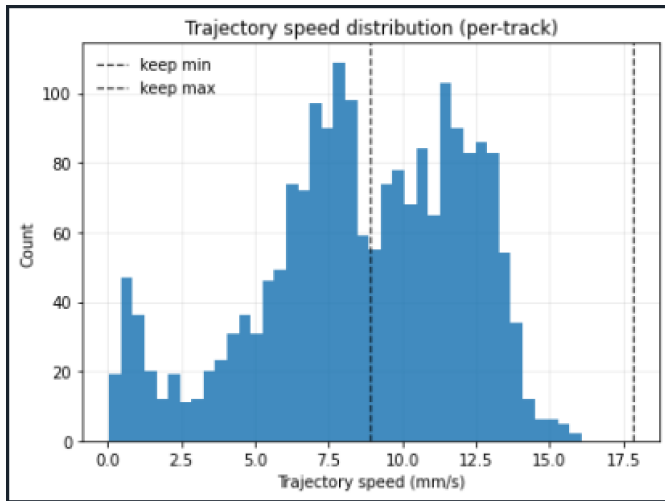
$D = 13\text{mm}$ (slit centre-to-centre separation)

$\lambda = 7.0073\text{mm}$ (wavelength)

This produced:



More plots of the double slit results:



Weekly Summary (Week 17)

- I was absent from the lab this week. My contributions were: independent measurement of Faraday wavelength from droplet photos (~ 4.5 mm); literature research on the dispersion relation from Bush (2015); discussions with Harry about the double-slit results.
- Harry's lab work: replaced oil (16/02/26), collected double-slit data at 70.5 Hz / 14.0–14.1 V with slit widths 14 mm ($d = 17$ mm) and 10 mm ($d = 13$ mm). Identified and reduced leftward trajectory bias by taping camera opening. Added stacked magnets and 15 ml additional oil.
- Automated droplet generation implemented: new Arduino code controls speaker drive and droplet launcher via relay, pausing between drops to let the wavefield reset. Circuit diagrams recorded.
- Filtered double-slit trajectories by speed (removed < 9 mm s^{-1}). Symmetrised angular distributions about the central peak to account for $\sim 10^\circ$ systematic bias.
- Fitted Faraday wavelength from single-slit Fraunhofer envelopes: 5.8 mm (12 mm slit), 8.3 mm (17 mm), 6.9 mm (22 mm), average $\lambda_F \approx 7.0 \pm 0.17$ mm. (Later superseded by dispersion relation value of 5.19 mm.)
- Compared double-slit data ($L = 10$ mm, $d = 13$ mm, $N = 529$) to theoretical $|\text{sinc} \times \cos|^2$ envelope using $\lambda = 7.0$ mm. Reasonable agreement after symmetrisation.

Plan going into Week 18

- Collect more double-slit trajectories with the second slit configuration to compare interference patterns.
- Confirm Faraday wavelength from the dispersion relation for comparison with the fitted values.
- Repeat single-slit measurements for slits with small trajectory counts to increase trajectory counts.

Success Criteria

- Double-slit data for both configurations with ≥ 500 trajectories each.
- Independent Faraday wavelength estimate from dispersion relation documented.

Week 18

27 February 2026

14:52

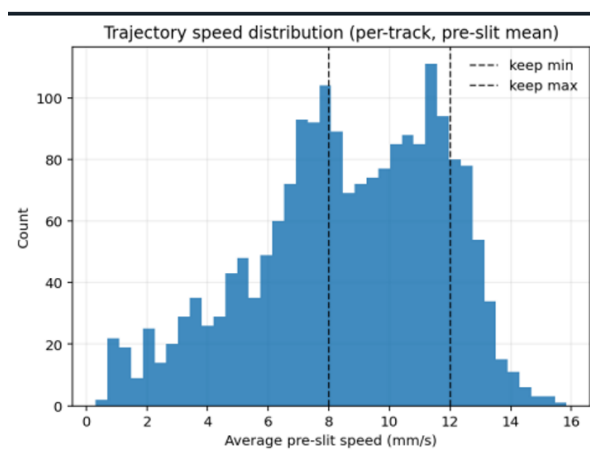
Date	Slit width (mm)	Frequency (Hz)	Amplitude V	Temperature (degrees Celsius)	Humidity	Oil height (mm)	Submerged height (mm)
23/02/26	10mm double (d=13mm)	70.5	14.0	24 - 29 (increase all day)	55.6%	5.5+0.5mm	0.7+0.5mm
24/02/26	17mm single	70.5	14.0	-	56.7%	5.5+0.5mm	0.7+0.5mm
26/02/26	22mm single	70.5	14	-	51.1%	5.5+0.5mm	0.7+0.5mm
26/02/26	10mm single	70.5	14	-	50.5%	5.5+0.5mm	0.7+0.5mm
27/02/26	10mm single	70.5	14	-	59.7%	5.5+0.5mm	0.7+0.5mm
27/02/26	12mm single	70.5	14	-	51.5	5.5+0.5mm	0.7+0.5mm

As we saw the temperature changing throughout the day we decided take recordings of the temperature every hour using a IR temperature gun. We had to let the bath heat up for an hour before we started taking readings.

Time	Temperature (degrees) 23/2	Temperature (degrees) 24/2	Temperature (degrees) 26/02	Temperature (degrees) 27/02
10:00	16.6	19.0	20.5	23.5
11:00	27.0	25.3	27.5	27.3
12:00	27.6	29.2	31.4	30.1
13:00	28.5	30.8	31.3	30.9
14:00	28.9	32.5	31.4	30.5
15:00	28.8	32.2	30.6	30.6
16:00	29.6	33.0	30.9	30.3

I was not present for the sessions at the start of the week (Monday 23/02 and Tuesday 24/02). The notes below for those days are based on my lab partner's lab book, the recorded tables, and later discussion of the results. I was present for the later sessions that week the 26th and 27th.

Like mentioned in the previous weeks my lab partners harry was talking about the speeds of the droplets varying massively and this was due to the droplet launcher not being able to produce droplets of one size. And we still conclude that we need this speed filtering as the speed has a massive effect on the extent to which they are diffracted. As if they move slower through the slit they will be diffracted more as will have a longer time interacting with the slits. This is why we need to adjust the range of speed we select for each different slit size. We also see a correlation in slit width and droplet speed. The larger the slit the larger the average velocity.



He then mentions about why larger droplets travel faster and if this has something to do with the wavelength of the faraday waves. He includes an equation from bushes paper which I also found the previous week. And found our faraday wavelength to be $5.193 \pm 0.05 \text{ mm}$ and this roughly aligns with what I found last week when absent so we agreed on that value.

$$\left(\frac{\omega}{2}\right)^2 = \omega_\omega^2(k_F) = \left(gk_F + \frac{\sigma k_F^3}{\rho}\right) \tanh(k_F H)$$

For a fixed driving frequency (and faraday frequency), gravitational acceleration, fluid density, surface tension and oil depth the faraday wavenumber and therefore wavelength remain unchanged.

With values of:

$$f = 70.5 \pm 0.0005 \text{ Hz}$$

$$\rho = 950 \pm 10 \text{ Kg m}^{-3}$$

$$\sigma = 20.3 \text{ mNm}^{-1} \text{ at } 22 \text{ degrees (online source)}$$

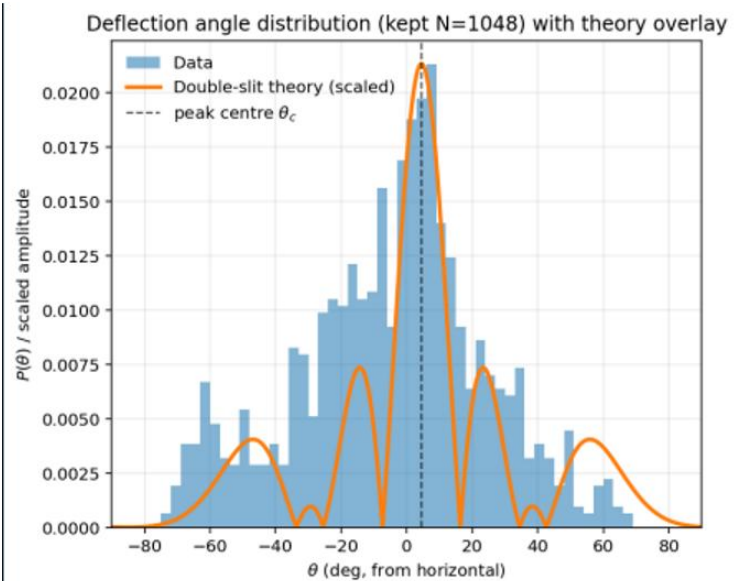
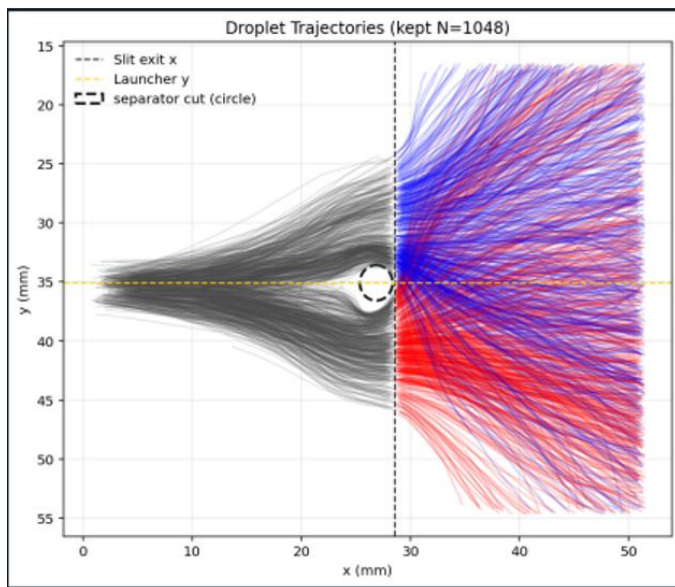
$$\frac{\partial \sigma}{\partial T} = -0.06 \text{ mNm}^{-1} \text{ K}^{-1}$$

Using this partial differential relationship the value of surface tension at 28 degrees is 19.94 mNm^{-1} .

This gave a value for $\lambda_F = 5.193 \pm 0.05 \text{ mm}$ which aligned roughly with the value found for the slit separation of 12mm.

Harry carried on looking at how droplet size was affecting speed. Larger droplets were generally seen to travel faster, because their greater mass and larger contact area transfer more momentum into the bath at each impact. Although bigger droplets should also experience more viscous drag, the increased propulsion from the wave field is the stronger effect, so the larger droplets end up moving with higher average speeds.

The further readings taken for the double slit ($d=13\text{mm}$, $L=10\text{mm}$) the results were more promising:



After collecting these results Harry mentions that these haven't been reflected which is what was done on the 2006 paper by Couder and even though there is a bias to the left hand side there peaks still align with the theoretical overlay. These were his words:

This angle deflection histogram has not been reflected about its peak and whilst there is still a slight bias towards the negative deflection angle the peaks on both side are aligning with the theoretical overlay. The speeds were filtered to only keep 8-12mm/s trajectories. More results could be taken.

The imbalance is still there however the data is approaching the theoretical distribution, with more data being collected the experimental values should mirror this distribution. Furthermore with both single and double slit patterns being produced, this aligns with John Bush's recent paper whereby disproving the fact that a purely classical system can't produce both single and double slit diffraction patterns.

Also since we had taken so many results for the double slits we went back and decided to spend the rest of our time taking more data for the single slits and making sure we could reproduce this data.

I came back in on Thursday and got a load of measurements for 22mm which took a while to get going because the system wasn't working again showing the fine tuning needed for this experiment, but after a few adjustments results were taken for the 22mm case. I then spent the Thursday evening and Friday morning collecting 10 mm to see if we could produce some results were in the Fresnel case we have 2 peaks.

The results taken were alright until we realised as you'll see next week that we may have had a slight bit too much oil in the system. We got some decent results but as you can see there are some trajectories which go over the slits and this is due to there being too much oil. This meant that there was still the slits there so a Fraunhofer case would be created but depending on the slit width. 10 mm was okay due to the slits being closer creating a better boundary effect. But there would be not much of a pattern in the Fresnel case.

Massimo wanted us to measure the speed of the wave created by the relay just before the droplet gets released. We took videos of it and were able to record the wave. It moved around 15mm in 10ms so very fast and because of this we think it has no effect on the droplet as it has gone before the droplet starts bouncing out at the launcher.

Weekly Summary (Week 18)

- Collected additional data across multiple slit widths: 10 mm double-slit ($d = 13$ mm) on 23/02, 17 mm single on 24/02, 22 mm and 10 mm single on 26/02, 10 mm and 12 mm single on 27/02. Oil height 5.5 ± 0.5 mm, submerged height 0.7 ± 0.5 mm.
- Monitored temperature hourly across four days using IR temperature gun. Observed significant variation ($16.6\text{--}33.0$ °C) requiring the bath to warm before data collection each morning.
- Confirmed speed-dependent diffraction: slower droplets interact longer with the slit and are diffracted more strongly. Speed filtering (slit-width-dependent ranges) is essential. Larger slits correlate with higher average droplet speeds.
- Calculated Faraday wavelength from the dispersion relation using $\rho = 950$ kg m⁻³, $\sigma = 19.94$ mN m⁻¹ (temperature-corrected to 28 °C), $f = 70.5$ Hz. Obtained $\lambda_F = 5.193 \pm 0.05$ mm, consistent with my independent estimate from Week 17.
- Harry investigated droplet size vs speed: larger droplets travel faster due to greater momentum transfer at each bounce, despite higher viscous drag.
- Double-slit results ($d = 13$ mm, $L = 10$ mm, $N = 1048$) showed peaks aligning with theoretical overlay even without symmetrisation, consistent with Couder and Fort (2006). Some residual leftward bias remains.
- Returned to single-slit data collection (22 mm, 10 mm) to increase statistics. Noted some trajectories passing over slits due to slightly excessive oil depth.
- Measured relay-generated wave speed at Massimo's request: ~ 15 mm in 10 ms. Too fast to affect the droplet, confirming it dissipates before the droplet begins walking.

Plan going into Week 19

1. Attempt to reproduce Fresnel near-field structure, for the 12 mm slit.
2. Collect remaining single-slit data to finalise all datasets before the end of term.

Success Criteria

- Faraday wavelength determined independently from the dispersion relation with uncertainty.
- Double-slit angular distribution showing peak structure consistent with theoretical predictions.

Week 19

02 March 2026 10:22

Write in report about how there is a bias to one side. It is showed in all of the results. Have spoken before about this and we think it is about the speaker probably.

Just spoke to Massimo and the 12mm results we just tried to reproduce we didn't get any distinctive Fresnel pattern like before. We actually see no patterns at all. A theory to why this could be is either due to the new set up of how the bath runs where it turns off and on after every drop. Or the more likely one is the oil depth. We see that there are a lot more droplets getting stuck on the submerged slit which we didn't see before and this could be due to the oil being 0.5mm deeper than before. Also when looking at the submerged slits there is a well around the edge of them and this well is less pronounced when the oil is deeper. So a thought is that the wells around the slits has a part in why these Fresnel patterns are forming. So tomorrow we are going to try and reduce the amount of oil and get it back to what it was last time and see if we can reproduce these Fresnel patterns with 12mm.

Date	Slit width (mm)	Frequency (Hz)	Amplitude V	Temperature (degrees Celsius)	Humidity	Oil height (mm)	Submerged height (mm)
02/03/26	10mm double (d=23mm)	70.5	14.0	~30C	49.9%	5.5 ±0.5mm	0.7±0.5mm
02/03/26	12mm	70.5	14.0	~30C	50.4%	5.5 ±0.5mm	0.7±0.5mm
03/03/26	12mm	70.5	14.0	-	46.7%	5±0.5mm	0.5±0.5mm
03/03/26	12 mm 2 Pipette more oil	70.5	14.0	-	48.3%	5±0.5mm	0.5±0.5mm
05/03/26	12 mm 1 Pipette more oil	70.5	14.0	-	51.7%	5±0.5mm	0.5±0.5mm
05/03/26	10mm	70.5	14.0	-	53.5%	5±0.5mm	0.5±0.5mm
06/03/26	12mm	70.5	14.0	-	46.3%	5±0.5mm	0.5±0.5mm

Time	Temperature (degrees) 3/3	Temperature (degrees) 5/3	Temperature (degrees) 6/3
10:00	22.6	22.6	22.9
11:00	28.9	26.1	24.2
12:00	30.5	27.8	28.7
13:00	31.1	28.6	29.4
14:00	28.4	28.7	30.4
15:00	29.2	28.5	29.2
16:00	28.9	29.2	29.4

We tried to repeat the 12mm single slits results Monday afternoon but was unsuccessful and found that there was no pattern happening in the Fresnel case. We thought that this could be due to their depth of the oil being too deep. It was an extra 0.5mm deep which meant the wells around the slits weren't pronounced as much as before. Today (Tuesday) we have reduced the depth of the oil to 5mm bath depth and 0.5mm slit depth. Hopefully this makes the slit wells more pronounced and this could be the cause for why the Fresnel pattern shows. Showing that the parameters have to be very finely tuned for reproducible results. Although the Fraunhofer case still showed but less distinctly as the first one. This is probably due to there being half the amount of readings.

Now the oil depth has been reduced you can see a much larger effect the slit wells are having. Also temperature of bath has hit 28C and more droplets are starting to go through the slits. Temperature has a big effect. Higher temperatures change the oil properties (surface tension, viscosity) which can shift the Faraday threshold and affect walking behaviour.

With the oil reduced we have found that the drops tend to get stuck in the wells of the slits meaning we are taking loads of trajectories and only a few are being selected. This shows how sensitive the height of the oil is to the experiment. Too much oil the slits aren't felt enough and too little oil the droplets do go through the slits properly. Saying this though, when filtering the slits which went through there is a clear sign the Fresnel pattern is coming back which backs our theory of the wells playing a significant part in the Fresnel case.

I have now added two 1ml pipettes worth of oil (15:24). Hopefully this will get us back to the goldilocks amount of oil. As seen in the Week 15 - Goldi set of graphs.

Unfortunately that did not work and I think it's starting to trend to too much oil again. On Thursday we will remove 1 pipette of oil and try again to reproduce these clean results.

We have come in this morning (Thursday) and turned on the experiment to get the oil heated up. We ran the 12mm slit all morning and got successful results. Apart from the fact they are very skewed to the right they give the correct readings at the goldilocks level. Which is what we were looking for. Although having a second look at it now this may not be the case. When looking at the original goldilocks 12mm trajectories there are 5 clear paths with 2 edge ones where as in the one done just now there are only three paths and clearly in a much smaller region. Also the Fraunhofer case seems to have not come through correctly. But then again the point source has been fired from is slightly below which could be causing this?

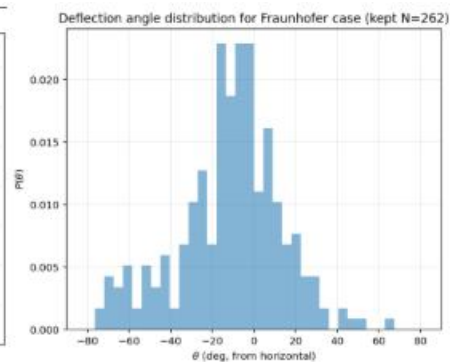
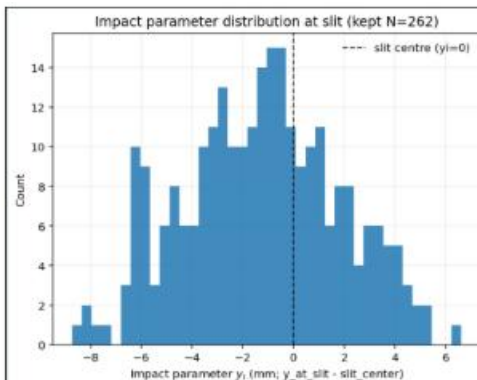
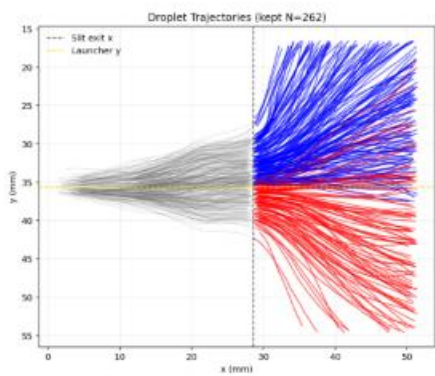
We have tried to collect some more 10mm data to see if we can get 2 peaks in the Fresnel case. We got these two peak which line up so in the Fresnel case and in the Fraunhofer case we get these two side lobes which are further out at a larger angle which is what you'd expect for a small slit width. The results obtained here support the oil was back at its Goldilocks level.

Final day of Labs and we are going to retake 12mm for the final time. We are going to make sure that the point source is perfectly lined up with the slit this time. Height of oil has stayed the same and we've found this is the perfect amount.

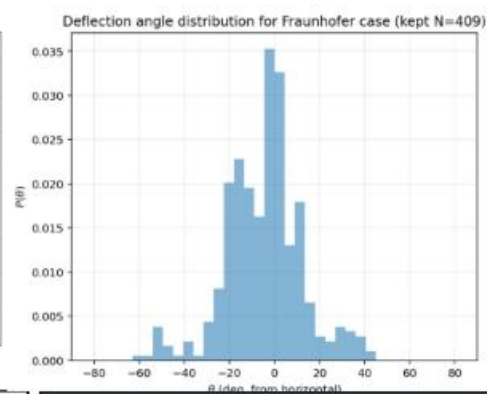
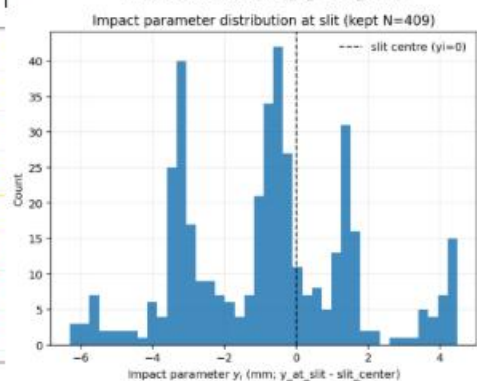
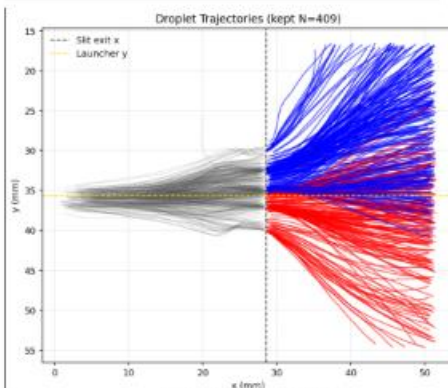
Next time we need to find a way to measure the height of the oil more precisely and also have something which keeps the temperature of the oil constant.

We spoke with Massimo about the Fresnel line up and we thought that the real width of the slits could be smaller than where the wavelengths are? So for example in the 12mm the outer bit of the slits can be ignored so the theoretical Fresnel case lines up with the experimental results and in the 17mm if it's does in a percentage basis than half of the outside would line up still so the 17mm theoretical case does line up with the experimental.

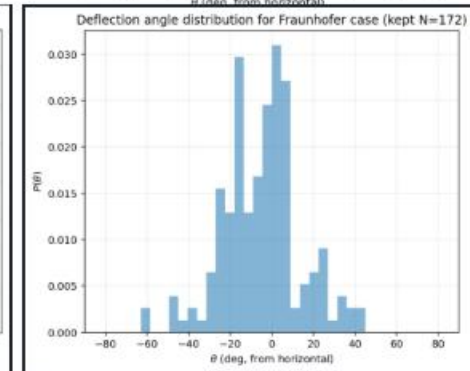
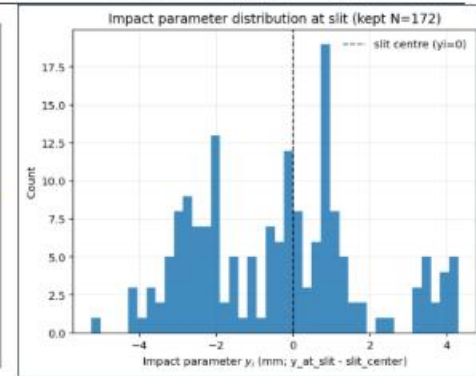
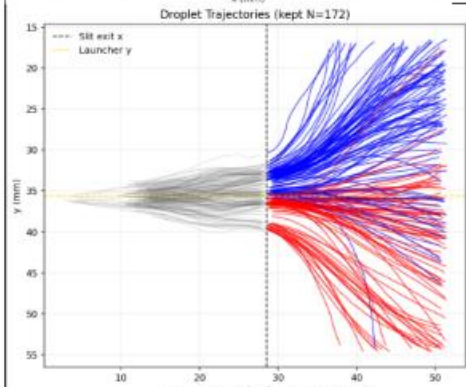
Too much oil:



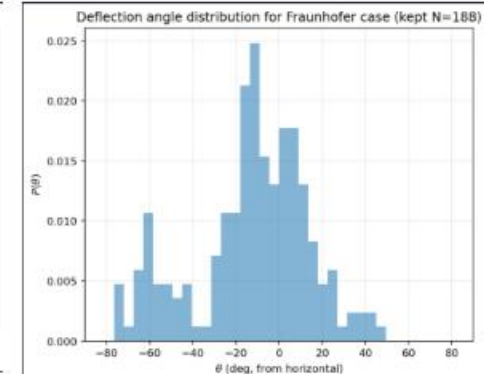
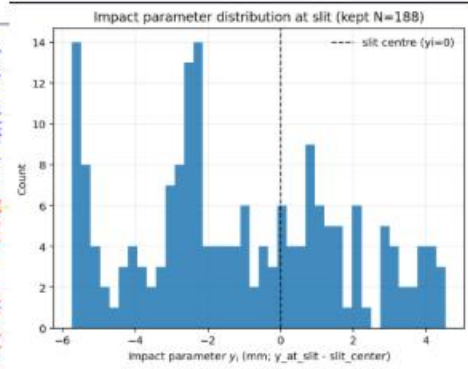
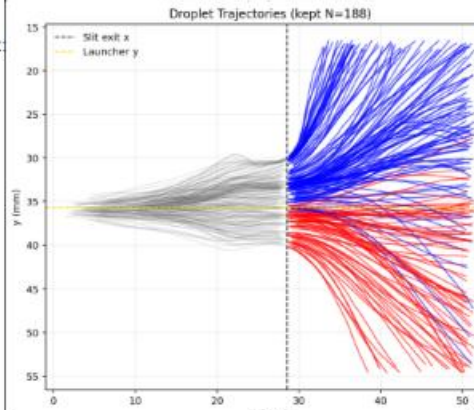
Week 15 - Goldi:



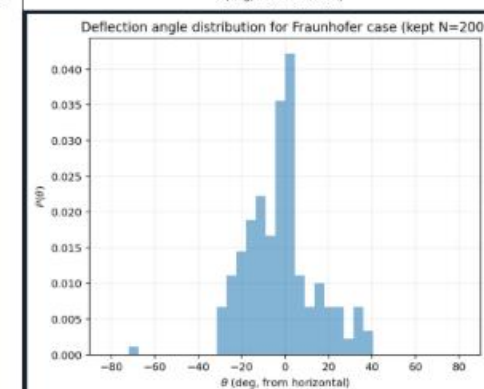
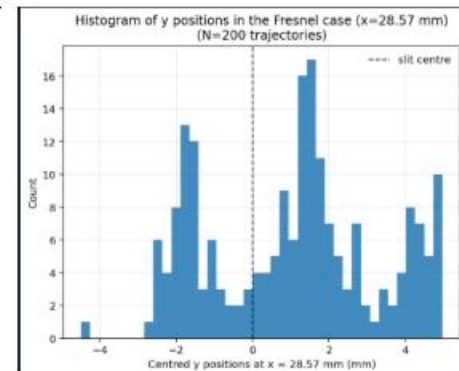
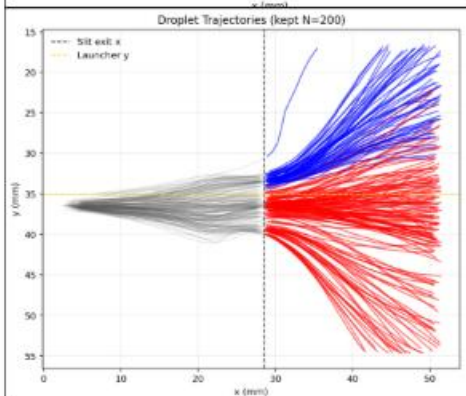
Too little:



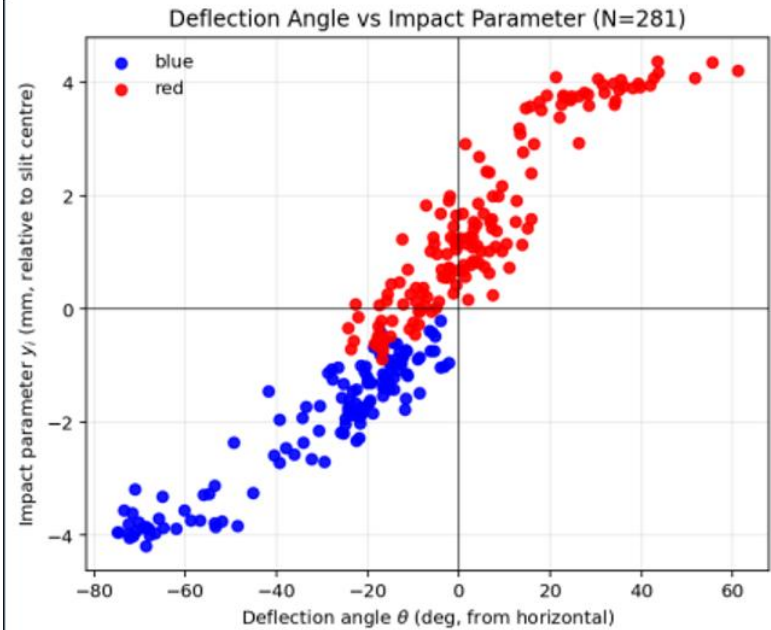
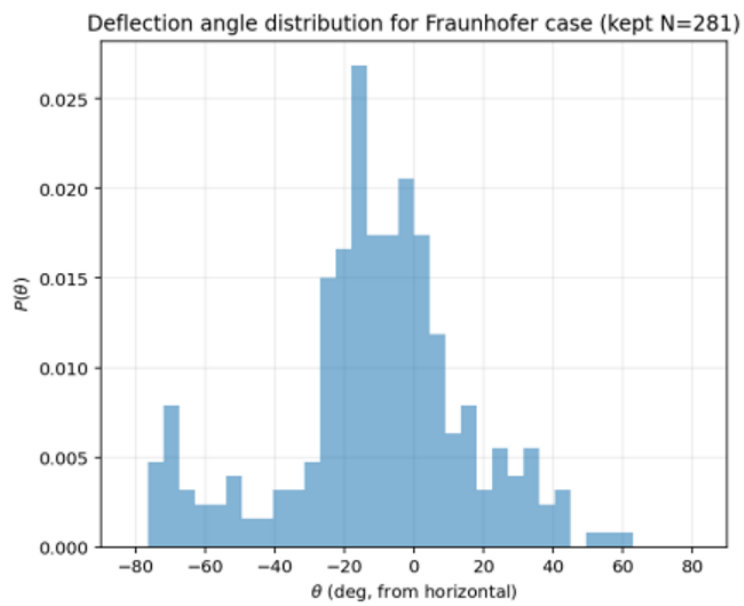
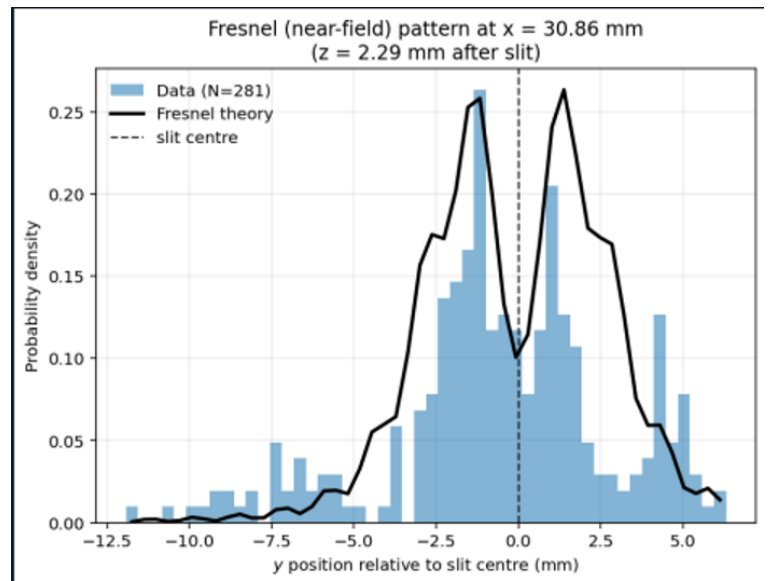
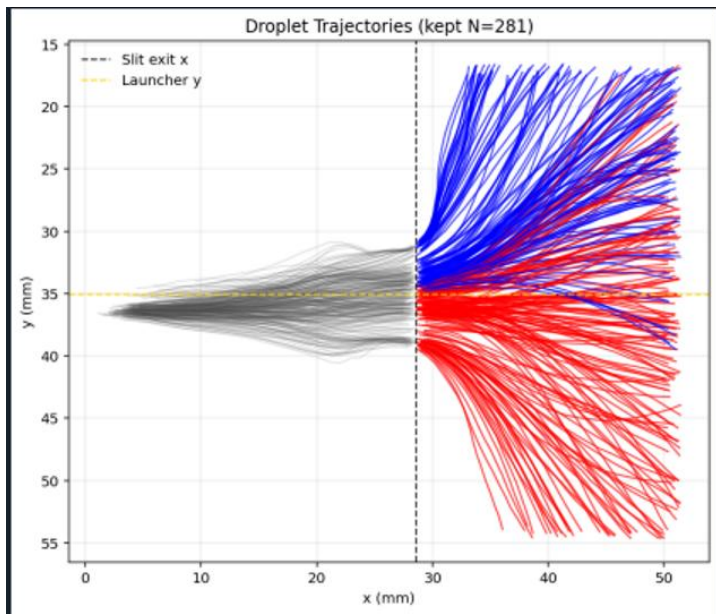
Too much 2



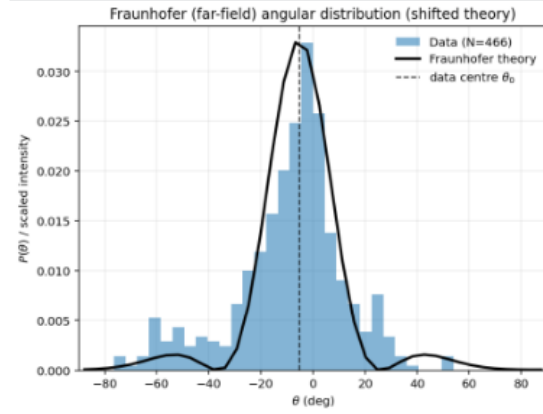
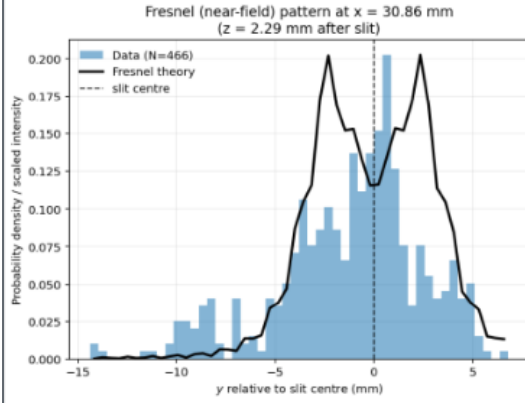
Goldi:



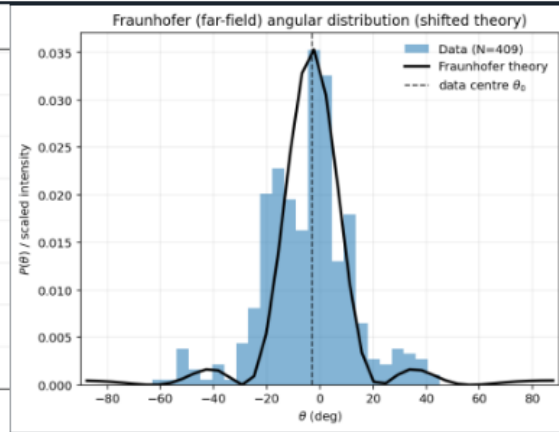
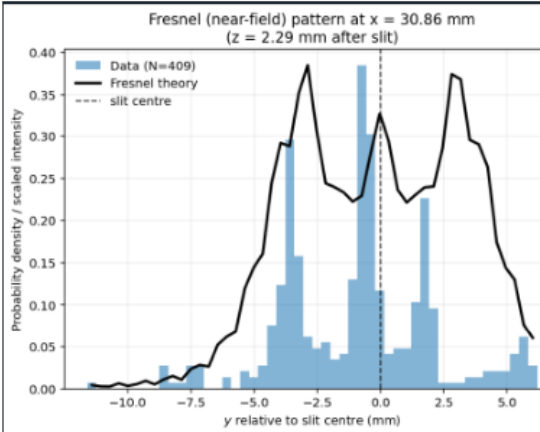
10 mm data:



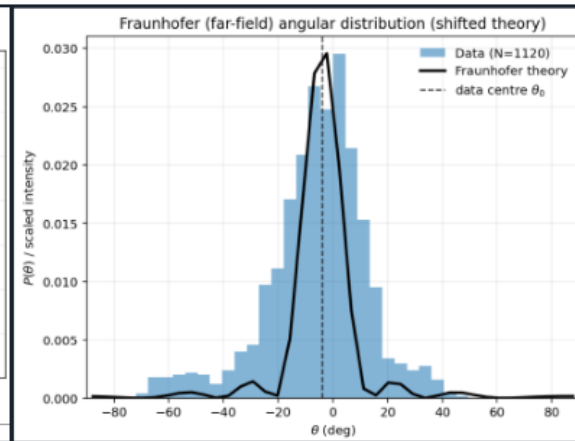
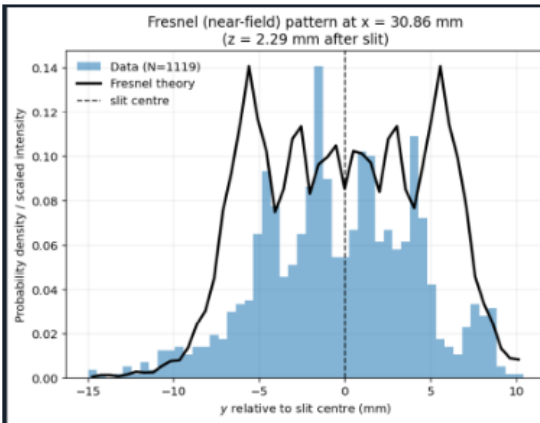
10 mm



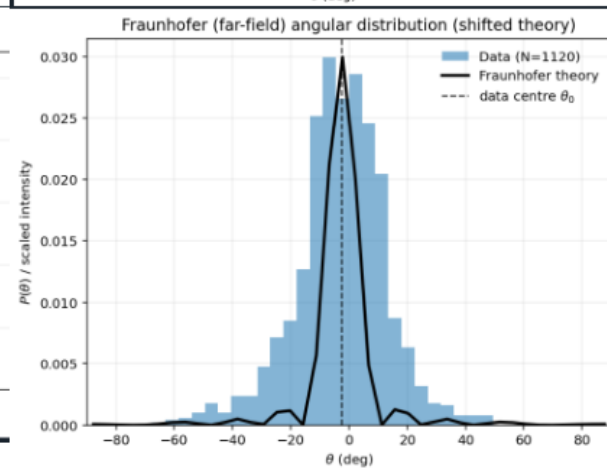
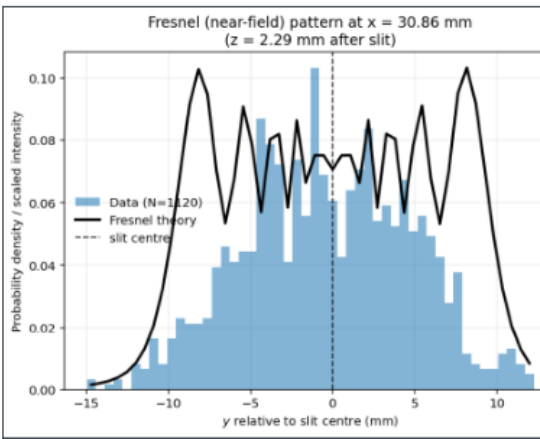
12 mm



17 mm



22 mm



Majority of slow particles are on the edge so can't make a complete call on the wavelengths as if the wavelength is related to the speed how can it have the same phase if it slows down it have a different wavelength.

At some point in QM if you have a small slit and make it smaller it won't go through. In reality the number of photons that go through. Particle slowing down when closing the slit. Could be due to the waves or speed of it.

Looking at the figures in the Darrow and Bush 2025 paper. The width of the deflection angle vs impact parameter shows this De Broglie mechanics line up. Also when looking at the trajectories you can see this cross over at nodes. This would be worth discussing in the report.

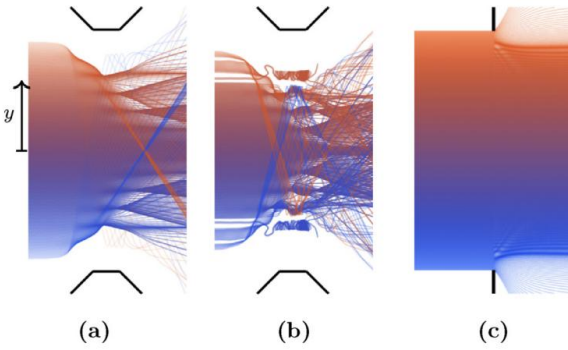


FIG. 2. Single-slit trajectories with slit width $w = 4.07\lambda_c$, initial momentum $p_0 = 0.3m\gamma$, 500 evenly spaced impact parameters $|y| < w/2$, and coupling constants (a) $b = 16.7$ and (b) $b = 25.0$. Trajectories cross in regular fold patterns, with spacing $0.28\lambda_c$ between folds in both simulations. (c) Bohmian trajectories deduced for matching slit width and wavelength. Notably, Bohmian trajectories do not cross one another in diffraction experiments [22,23].

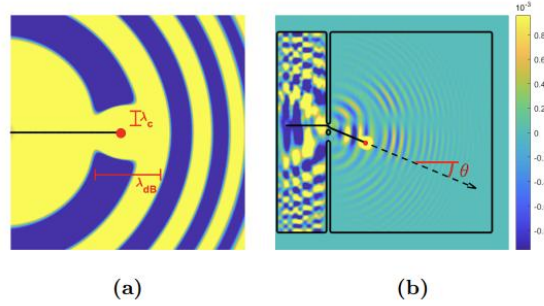


FIG. 1. (a) A close-up of the free particle in a steady state, reproduced from Ref. [9]. The image is color-coded according to the signed amplitude ϕ of the pilot wave. The particle radiates energy only when it accelerates, but is always accompanied by a Compton-scale wave packet and a quasimonochromatic pilot wave field with the de Broglie wavelength. (b) A trajectory in our double-slit experiment shows the particle exiting the slit at a diffraction angle θ . The standing wave pattern apparent behind the slit only materializes once the particle has exited the slit, so does not influence its dynamics.

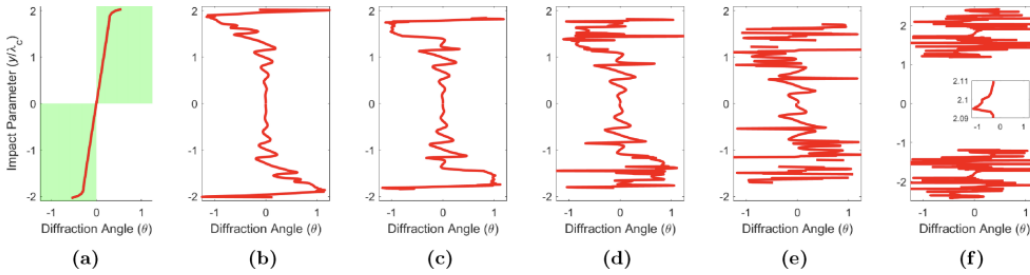


FIG. 5. Diffraction maps relate incoming impact parameter y_{in} to far-field diffraction angle θ_{out} , measured in radians. (a) Bohmian diffraction map corresponding to Fig. 2(c). All trajectories are “surreal,” and so do not cross the system’s centerline; Bohmian trajectories thus satisfy the first criterion of Proposition A1, allowing them to exhibit Fraunhofer diffraction. (b)–(e) Diffraction maps corresponding to our single-slit simulations, with slit width $w_1 = 4.07\lambda_c$, initial momentum $p_0 = 0.3m\gamma$, and coupling strengths $b = 12.5, 16.7, 20.9, 25.0$. As is evident in Figs. 2(a) and 2(b), as b increases, regular fold patterns are replaced by chaotic behavior; our system thus satisfies the second criterion of Proposition A1. (f) A diffraction map corresponding to our double-slit experiment in Fig. 3(d), with slit width $w_2 = 2.03\lambda_c = w_1/2$, slit separation $d = 3.66\lambda_c$, initial momentum $0.3m\gamma$, and coupling strength $b = 25.0$. A cutout at $2.09 < y < 2.11$ reveals that the map is continuous but not smooth. Since it is not smooth, our model is able to exhibit a smooth (e.g., Fraunhofer) diffraction pattern.

Weekly Summary (Week 19)

- Attempted to reproduce the 12 mm Fresnel pattern from Week 15 but initially unsuccessful. Identified oil depth (0.5 mm deeper than Week 15) as the most likely cause: deeper oil reduces the wells around the submerged slits, weakening the boundary condition that produces the Fresnel structure.
- Systematically varied oil depth over the week: too much oil (N=262, no Fresnel pattern), Week 15 Goldilocks reference (N=409, clear pattern), too little oil (N=172, droplets stuck on slits), too much again (N=188), and final Goldilocks attempt (N=200, Fresnel pattern recovered). Documented with comparative trajectory/impact parameter/Fraunhofer plots for each condition.
- Collected 10 mm single-slit data showing two-peak structure in both Fresnel and Fraunhofer regimes, consistent with expectations for a small slit width.
- Collected additional double-slit data (10 mm, $d = 23$ mm) on 02/03.
- Continued hourly temperature monitoring (3/3, 5/3, 6/3). Morning temperatures 22–23 °C rising to ~30 °C by midday.
- Discussed interpretation with Massimo: the effective slit width seen by the droplet may be smaller than the geometric width, which would explain why the Fresnel nodal positions align with theory even when the outer region of the slit is less active. Also discussed speed-dependent wavelength and de Broglie-style trajectory crossing from Darrow and Bush (2025).
- Noted persistent directional bias across all results, likely due to the speaker. Identified as a systematic effect to discuss in the report.
- Identified future improvements needed: more precise oil depth measurement and temperature control.

Plan going into report writing

1. Write up a plan for the report using the Lab book as guidance.

Success Criteria

- All trajectory CSVs organised and cross-referenced with experimental parameters.
- Clear decision on which datasets are primary (report) vs supplementary (appendix).

Link to GitHub with all Code, data files, and CAD files:

<https://github.com/JacobBrooke1/walking-droplet-diffraction>